

Approximate Full-Adders: A Comprehensive Analysis

Ettore Napoli¹, Senior Member, IEEE, Efstratios Zacharelos¹, Member, IEEE, Antonio G. M. Strollo², Senior Member, IEEE, and Gennaro Di Meo²

¹Department of Information and Electrical Engineering and Applied Mathematics in the University of Salerno, Fisciano 84084, Italy

²Department of Electrical Engineering and Information Technology, Federico II, Naples 80125, Italy

Corresponding author: E. Zacharelos (e-mail: ezacharelos@unisa.it).



This work has been funded by the NextGenerationEU, PNRR M6C2 2.1.
Grant: D43C22005080006, Project title: CARE-MODE

ABSTRACT Approximate computing is a technique that sacrifices the accuracy of the result for an advantage in terms of power, area, and speed. It is useful for error-tolerant applications such as image and video processing. Many approximate arithmetic circuits can be devised using approximate versions of the basic binary full adder that simply adds three bits to generate carry and sum. Therefore, the approximate full adder has been the subject of extensive investigation in recent years. In this paper we present a comprehensive analysis of approximate full adders, by synthesizing in a FinFET 14 nm technology the whole set of possible designs, achievable by modifying one or more entries of the full adder truth-table. Our exhaustive analysis shows that only a few out of the many thousands of synthesized circuits perform reasonably well as approximate full adders. Our analysis re-discovers the approximate full adders already proposed in the literature and identifies some new ones. Examples of using the newly discovered approximate full adders in typical error resilient applications are provided, showing the performance and the usefulness of approximate arithmetic circuit designs in the real-world.

INDEX TERMS Approximate computing, approximate full adders, binary adders, binary multipliers, digital arithmetic.

I. INTRODUCTION

Approximate computing in digital design attempts to relax the requisite of exact computations to gain substantial performance improvement in terms of power, speed and area. Designers have always tried to minimize resources requirements by picking the smallest possible bit-width, and by approximating complex operations [1]. However, the topic of systematically studying approximate circuits to be used in error-resilient applications like image processing, multimedia processing, data mining and machine learning [2],[3], possibly implemented in edge or embedded systems that are characterized by severe energy and speed constraints, is relatively recent, [4]-[9].

Approximate computing has been applied to binary multipliers [10],[11], binary dividers [12], MAC units [13], and entire systems [13],[14]. Since the full adder (FA) is a ubiquitous building block for arithmetic circuits (from adders to multipliers, dividers, etc.), several papers have proposed various versions of an approximate full adder (named approxFA in the following), and employed them in

many application contexts [14]-[60].

In [16],[17] five approximate full adders (approxFAs) are proposed and designed at transistor level in 90 nm CMOS technology. The circuits are compared in terms of error and power dissipation and their performance is compared against simple output truncation. In addition, in [17] an algorithm for the optimal use of the approxFA as a function of the allowed error is proposed. In [18] three approxFAs designed at transistor level by using pass transistors and XOR/XNOR gates are proposed. In [19] two energy-efficient transistor-level approxFAs are proposed. These designs use two transistors as switches driven by an enable signal to activate the approximate mode at runtime. Authors in [20] designed three approxFAs at transistor level, keeping the sum exact, and the carry equal to each input, respectively. Ref. [10] presents an approximate adder that sums generate and propagate signals to pack the input bits in one of the operands and then approximate the sum. The adder is exploited in binary multiplication with performances reported for both FPGA and ASIC in a 28 nm technology. Ref. [21] and [22]

propose three and two inexact FA cells, respectively. The electrical performances are given for a 45 nm technology. The approxFAs are tested in image processing applications.

A survey of the approximate computing techniques until 2016 is presented in [23] where the approxFAs proposed in [16],[17] are exploited to build multi-bit adders.

In [24] four approxFAs, one of which already discussed in [21], are proposed and used for the design of approximate adders with a bit size up to 128 bits. The results report up to 98% power dissipation reduction. In [25] four approxFAs are proposed and implemented in a 7 nm FINFET technology. The focus is on transistor level implementation and voltage scaling. The circuits are compared against the exact and approximate designs taken from [18] and [26]. Three approximate full adders synthesized for a 180 nm technology are proposed in [27], while ref. [28] exploits nine previously proposed approxFAs synthesized using the ABC tool and a 180 nm standard cell library. The approxFAs are exploited to design ripple carry adders (RCAs) by means of a genetic algorithm.

ApproxFAs are also proposed for adiabatic logic applications in [29] where two novel designs are proposed.

In [30] four transistor level approximations of the full adder are tested against an exact MUX-based full adder design. The paper shows that the approximation does not always result in a reduction of the power dissipation.

The authors in [31] focus on the process of carry generation to create an approxFA that improves the delay of the carry propagation path. Three approxFAs with low overhead are presented in [32] in which one of the inputs is used as a clock control signal to enable the transmission gate of the adder.

Ref. [33] proposes an approxFA exploited in a 4-bit multiplier. Approximate full adders are also tested in a 32 nm CNFET technology in [34],[35] where new versions and logic styles of the circuits are proposed. In [36] the design of the binary multiplier is addressed. The sum of certain FAs, employed to reduce the partial product matrix, is approximated with OR gates, obtaining good electrical performance. In [37] approximate half and full adders are proposed for the design of approximate recursive multipliers tested in image processing applications. In [38] a new family of approximate compressors, including approximate half and full adders, aiming to reduce the error probability and the average error is presented. Authors in [39], propose five approxFA designs, and use them to generate various 4-2 approximate compressors with carry-in and carry-out, that are ultimately used in an 8-bit Dadda multiplier.

Memristor-based approxFAs, based on the IMPLY logic are presented in papers [40] and [41]. The approxFA proposed in [40] is used in a hybrid, semi-serial structure while the four designs proposed in [41] are used in a fully serial architecture, aiming to minimize the silicon footprint.

The so called "Almost Full Adder" is developed in [44] by AND-ing two inputs for the carry and keeping the sum exact.

The impact of the design is investigated as it is placed in different stages of the Dadda multiplier. In [45] the authors develop approximate full and half adder designs as well as an approximate 15-4 compressor and use them in a 16-bit binary multiplier. A Kogge Stone Adder and an approxFA are employed in [46] to build a 16-bit Wallace multiplier. Four approxFAs are derived by simplifying a conventional mirror adder in [47], and then used for multipliers.

The authors in [48] propose various types of approxFAs, including NAND-based, carry-centered, and majority logic based adders, which are evaluated for their performance features and used in RCAs. Some of the same circuits are used in [49] to build 16-bit and 32-bit adders.

Lower Part Approximate and Carry-based Approximate Adders are investigated and proposed in [50]. Four more approxFAs are in [51]. The authors in [52] design two approxFAs at transistor level using 12, and 10 transistors.

The long list of papers that propose approximate full adders reflects the importance of the topic that is explored in different ways and contexts. An approxFA, can be designed using techniques based on logic simplification, logic manipulation, and transistor level design, but also by applying system level action, such as voltage overscaling as in [53]. Furthermore, the approxFAs can be exploited as building blocks for the design of more complex arithmetic units such as the higher-order approximate compressors that are key components in approximate multipliers, [54]-[61].

However, to the best of the authors' knowledge, none of the scientific contributions proposed to date reached a definite result regarding the whole set of possible approximate full adders, nor suggested which are the best performing approximate full adders.

To further complicate matters, it should be noted that in some cases, the authors use design styles for their approxFAs that are rarely employed in practice (such as pass-transistor structures or CNFET devices). This makes it difficult to assess the potential benefits associated with the proposed circuits in a standard cell design. Moreover, a proposed approxFA may be re-proposed in subsequent papers under a different name, possibly with a different design style. As an example, the approxFAs IMPACT1 [16], APP3 [17], CBA1 [24], AMA1 [25], aAMA3 [30], HIAFA [40] all have the same truth-table; similar observation holds for APP2 [17], InXA3 [21], AXMFA [22], CBA4 [24], TCAA [29], aAMA2 [30], CMIFA3 [35], MAA3 [47], AA5 [48], MLFA1 [49] that have identical logic behavior.

In this paper we perform a comprehensive analysis of approximate full adders, by exploring the entire design space of approximate circuits that can be generated through alterations to the entries of the full adder truth table. To obtain results as consistent as possible with the digital design techniques used in practice, all circuits are implemented using a commercial FinFET 14 nm standard cell library.

Despite the huge search space considered, our investigation shows that only a few of the synthesized

TABLE I
TRUTH TABLE OF THE EXACT FULL ADDER

i	c_{in}	a	b	c_o	s	S
0	0	0	0	0	0	0
1	0	0	1	0	1	1
2	0	1	0	0	1	1
3	0	1	1	1	0	2
4	1	0	0	0	1	1
5	1	0	1	1	0	2
6	1	1	0	1	0	2
7	1	1	1	1	1	3

TABLE II
TRUTH TABLE OF AN APPROXIMATE FA

i	c_{in}	a	b	c_o	s	S_{app}	$Err_i = S - S_{app}$	$RErr_i$
0	0	0	0	0	0	0	0	-
1	0	0	1	0	1	1	0	0
2	0	1	0	0	1	1	0	0
3	0	1	1	1	1	3	-1	-1/2
4	1	0	0	1	1	3	-2	-2
5	1	0	1	1	0	2	0	0
6	1	1	0	1	0	2	0	0
7	1	1	1	1	0	2	1	1/3

circuits perform reasonably well as approximate full adders. Moreover, we show that several approxFAs previously proposed in the literature deliver poor performance when implemented using a practical standard cell library. Ultimately, two sets of eight and nine optimal approximate full adders are found. While some of these optimal designs correspond to previously proposed circuits, the rest are new.

The final section of the paper shows examples of using the approxFAs in typical error resilient applications such as image addition, image blurring, and a dense artificial neural network.

The contributions of the paper are summarized as follows:

- We perform for the first time a comprehensive analysis of all the approximate circuits that can be generated by modifying the truth table of a binary full adder. The explored design space is composed of a total of 13,055 different circuits and includes all the previous approxFAs proposed in literature so far.
- We extract the optimal approxFAs on the Pareto front in the area-error and in the power-error planes.
- We show that, despite the huge available design space, only a bunch of circuits are Pareto optimal. While some of these optimal designs correspond to previously proposed circuits, others are new.
- We present results for error resilient applications demonstrating that the proposed set of Pareto optimal circuits consistently provides improved results when compared to the state of the art and can be profitably used in practical applications.
- For each of the three error resilient applications a single arithmetic unit is approximated (the carry ripple adder, the binary multiplier, or the multiply accumulate unit). Hence, the results are a precious hint of the electrical performance and the quality level that the proposed approxFAs can provide in different scenarios.

The paper is organized as follows. In Section II the search space and the design technique for the approximate FA is detailed. The results of the syntheses are discussed in Section III where the best trade-offs between error and electrical performance are identified. The applications are discussed in Section IV. The conclusions are in Section V.

i: 7 6 5 4 3 2 1 0 7 6 5 4 3 2 1 0
 1 1 1 1 1 0 0 0 0 0 0 1 1 1 1 0
 c_o s

FIGURE 1. Binary strings of c_o and s for the TT in Table II. ID = F81E.

II. DESIGN OF APPROXIMATE FULL ADDERS

The truth table of the exact full adder is shown in Table I. The inputs are a , b , c_{in} , while the sum and carry outputs are indicated as s and c_o , respectively. The arithmetic sum of the inputs is indicated as S and each entry of the truth table is numbered with the index i , ranging from 0 (when $\{c_{in}, a, b\} = \{0, 0, 0\}$) to 7 (for $\{c_{in}, a, b\} = \{1, 1, 1\}$). The logical equations of the full adder are:

$$\begin{aligned} s &= a \oplus b \oplus c_{in} \\ c_o &= ab + ac_{in} + bc_{in} \end{aligned} \quad (1)$$

We can modify (possibly simplifying) the expressions of s and c_o , by altering (i.e. complementing) some of the entries of the truth table. As an example, Table II shows a possible truth table of an approxFA with two modifications to the column of s (in the rows $i=3$ and $i=7$), and one modification to the column of c_o (in the row $i=4$). These modifications alter the logical equations as follows:

$$\begin{aligned} s' &= (a + b) \oplus c_{in} \\ c_o' &= c_{in} + ab \end{aligned} \quad (2)$$

Having two outputs (s and c_o) and eight entries in the truth table, we obtain a total of $2^{16}-1=65,535$ viable modifications, corresponding to the entire space of the possible approximate full adders that can be designed. Please note that this huge search space includes trivial designs in which both s and c_o are fixed to either 0 or 1, as well as all the approximate full adders previously proposed in cited references [9], [10], [16]-[22], [24], [25], [27], [29]-[37], [39]-[42], [44]-[52].

Each approxFA can be identified by a unique 4-digits hexadecimal number, named ID in the following, that encodes the binary strings of c_o and s . As an example, Fig. 1 shows the concatenated binary strings of c_o and s for the approxFA of Table II “1111 1000 0001 1110” whose ID is F81E, while the ID of the exact FA, whose concatenated binary strings are “1110 1000 1001 0110”, is E896.

TABLE III
APPROXIMATE FA WITH A TT EQUIVALENT TO TABLE II

i	c_{in}	a	b	c_o	s	S_{app}	$Err_i = S - S_{app}$	$RErr_i$
0	0	0	0	0	0	0	0	-
1	0	0	1	0	1	1	0	0
2	0	1	0	1	1	3	-2	-2
3	0	1	1	1	0	2	0	0
4	1	0	0	0	1	1	0	0
5	1	0	1	1	1	3	-1	-1/2
6	1	1	0	1	0	2	0	0
7	1	1	1	1	0	2	1	1/3

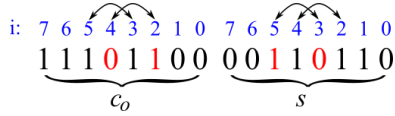


FIGURE 2. Binary strings of c_o and s for the TT in Table III. ID = EC36

A. REDUCING THE SEARCH SPACE

A full adder computes the sum of the three input bits c_{in} , a , b and encodes the result in the two outputs c_o and s . The input bits of a full adder are exchangeable due to the commutative property of the addition. In other words, the input bits can be swapped without changing the functionality and logic implementation of the circuit. This property can be exploited to reduce the search space.

Definition: Equivalent truth tables

Two different truth tables (TTs) of an approximate full adder are defined as equivalent if a permutation of the names of the inputs transforms one TT into the other.

Table III shows an example of a TT equivalent to the one of Table II, obtained by swapping c_{in} and a . The logical equations corresponding to Table III are:

$$\begin{aligned} s'' &= (c_{in} + b) \oplus a \\ c_o'' &= a + c_{in}b \end{aligned} \quad (3)$$

If the input bits of the TT in Table III are renamed so that $\{c_{in} \Rightarrow a, a \Rightarrow c_{in}\}$ and the input combinations are reordered as in Table II, the two TTs in Table II and in Table III coincide. It is important to notice that both truth tables represent the same synthesized circuit, with the sole distinction being that the input port names have been permuted.

Since the equivalent TTs are obtained by permutating the names of the input bits, let us consider all the possible permutations. The original sequence is: $\{c_{in}, a, b\}$ while the remaining five permutations are: $\{c_{in}, b, a\}$, $\{a, c_{in}, b\}$, $\{a, b, c_{in}\}$, $\{b, a, c_{in}\}$, and $\{b, c_{in}, a\}$.

This means that, given a TT and the associated synthesized circuit, the maximum number of equivalent TTs (resulting in the same synthesized circuits) is six.

It is worth noting that in a few cases some of the permutations result in the same TT (that is, the TT is invariant to the permutation). As an example, the TT composed by all '0' (or all '1') is invariant under all the permutations.

Algorithm 1 Derivation of the complete list of approximate FAs.

```

1: initialize analyzed_TT_list;
2: for ID=(0..FFFF) except E896
3:   if not (ID in analyzed_TT_list)
4:     create_equivalent_TTs;
5:     add {ID, equivalent_TTs}
6:     to analyzed_TT_list;
7:     generate_and_save_HDL (ID);
8:   end if
9: end for

```

B. SEARCH ALGORITHM

Given a TT, the equivalent TTs can be obtained by swapping some bits in its identifier ID. For example, the permutation $\{a, c_{in}, b\}$ of the TT in Table II corresponds to swapping the ID bits as in Fig. 2. Based on this observation, we have devised the search algorithm reported above.

Initially, an empty list of analyzed TTs is defined. Then a loop on the possible approxFA IDs, except for the ID of the exact FA, is run. For each ID, if not in the list of analyzed TTs, the set of the ID of equivalent TTs is generated. This step is conducted by properly swapping some of ID bits, depending on the considered permutation. The list of analyzed TTs is then updated and the Hardware Description Language (HDL) code corresponding to the TT is generated.

After the equivalent designs are considered, the number of different approxFAs is reduced to 13,055 (from 65,535). It can be demonstrated that the result is obtained by summing: 8,960 circuits that have six equivalent TTs (the TT changes for every permutation listed above); 3,840 circuits that have three equivalent TTs (the TT does not change if two particular bits are switched hence they change only for three of the six permutations listed above); 255 circuits that do not have equivalent TTs (the TT is invariant to every permutation).

The number is in the range of what is treatable with modern design tools and a workstation. In this paper the entire design space of the approxFAs has been tested to determine the electrical and error performance. For the sake of completeness, the exact FA is also synthesized and added to the list of approxFAs as a reference design.

C. ERROR METRICS

The error of the approxFAs depends on the entries of the truth table that differ from the truth table of the exact FA.

Let us define for each of the eight entries of the truth table of the approxFA:

$$S_{app} = 2 \cdot c_o + s \quad (4)$$

and the error with respect to the exact FA as:

$$Err_i = S - S_{app} \quad (5)$$

For each entry of the truth table, Err_i indicates the negative or positive error caused using the approxFA. An integral error metric considered in this paper is the sum of absolute errors defined as:

$$SAE = \sum_{i=0}^7 |Err_i| \quad (6)$$

The SAE value of an approxFA ranges from 0, for the exact FA, to 18 for the approxFA that has all the entries negated with respect to the exact FA. As an example, the SAE of the approxFAs in Tables II and III is 4.

The relative error of an approxFA, for each entry whose index is not 0 is defined as:

$$RErr_i = 1 - S_{app}/S \quad \forall i = \{1, \dots, 7\} \quad (7)$$

The mean absolute relative error is another important metric considered in this paper for the selection of the optimal approxFAs, and it is defined as the average of the absolute $RErr_i$ values:

$$MRED = \frac{1}{7} \sum_{i=1}^7 |RErr_i| \quad (8)$$

As an example, the $MRED$ of the approxFA in Table II and in Table III is $(|-2| + |-1/2| + |1/3|)/7 = 17/42 \approx 0.405$.

Please note that the $MRED$ does not consider the approxFA behavior when the input is $\{0,0,0\}$, since an error for this input value would result in an infinite relative error. Thus, approxFAs with an error when the input is $\{0,0,0\}$ could have $MRED=0$ while still producing an error.

The error probability represents the likelihood of producing an incorrect output, independently from the magnitude of the error. For the FA that has eight entries in its truth table, it is defined as:

$$PE = \frac{1}{8} \sum_{i=0}^7 (1 \text{ if } Err_i \neq 0 \text{ else } 0) \quad (9)$$

The mean error is the sum of the errors of all the entries in the truth table of the FA, divided by the number of entries. It is useful for some niche applications in which it is possible to compensate for the accumulated errors in subsequent steps. It is defined as:

$$ME = \frac{1}{8} \sum_{i=0}^7 Err_i \quad (10)$$

The maximum absolute error is useful when the design requires particular attention to the worst-case scenario and is simply defined as:

$$MaxAE = \max(|Err_i|) \quad i = \{0, \dots, 7\} \quad (11)$$

PE for the approxFA in Table II and in Table III, that has error in three entries, is $3/8=0.375$ while ME is $(-2-1+1)/8=-0.25$ and $MaxAE$ is 2.

In this paper the selection of approxFA performances from the error point of view has been based on the SAE and $MRED$ error metrics. As a matter of fact, these error metrics describe the overall behavior of a design, regardless of any particular application and are commonly used in the field, [37], [42].

D. ELECTRICAL PERFORMANCES

The entire space of approxFAs, with the addition of the exact full adder for a total of 13,056 designs, has been synthesized in a 14 nm FinFET standard cell technology (0.8V supply voltage, regular V_T) provided by Global Foundries. The synthesis is conducted with the Cadence Genus software executing 'syn_generic' followed by 'syn_map' both with effort high and 'syn_opt'. The timing constraint is set to the value of 1 ns to enhance the low power synthesis of the circuits. Power dissipation is computed by simulating the final netlist with outputs loaded with a NAND2X cell, stimulated with 10^5 uniformly distributed random inputs to obtain the switching activity of each node. The syntheses were performed with a workstation equipped with a 72-cores Xeon CPU E5 v3 2.3GHz, with 132GB RAM. Each synthesis takes about 20s, so that the total CPU-time to perform all the syntheses is in the order of 70 hours.

The electrical performance in this paper is measured in terms of silicon area occupation due to the cells, power dissipation, and maximum delay.

The exact full adder, synthesized with timing constraint of 1 ns, occupies $0.871\mu m^2$, has a power dissipation of $2.977\mu W@1GHz$, and a maximum delay of 74ps.

III. APPROXIMATE FULL ADDER RESULTS AND COMPARISON WITH THE STATE OF THE ART

A. APPROXIMATE FA PERFORMANCES

The implementation results are shown in Figs. 3, 4 where each dot represents a synthesized approxFA. In Fig. 3(a) the abscissa of each dot is the SAE of the approxFA, and the ordinate is its silicon area; Fig. 3(b) considers as ordinate the power dissipation of the approxFAs. In Fig. 4(a)-4(b) the quantity on the abscissa is the $MRED$.

The exact FA, highlighted with a red circle in Figs. 3-4, is the only circuit showing zero values for both SAE and $MRED$. Other points with $MRED=0$ correspond to circuits that exhibit erroneous behavior when all inputs are low, a case that this error metric fails to capture. The reader can also notice that there is just one point that provides $SAE=18$, corresponding to the approxFA that negates all the outputs of the exact FA.

The plots in Figs. 3,4 show limit cases with zero area or zero power dissipation. The approxFA designs with zero area correspond to trivial circuits in which the outputs are stuck to zero or one (for instance: $s=1$, $c_o=0$) or are directly connected to the inputs (for example: $s=a$, $c_o=c_{in}$). In such cases the synthesized approxFAs do not use any logic thus resulting in zero area circuits. The approxFAs with zero power dissipation correspond to the subset of zero area circuits in which the outputs are stuck to zero or one and do not dissipate any power since they do not have output changes thus causing no power dissipation on the load capacitance.

Figs. 3 and 4 reveal that within the approxFAs design

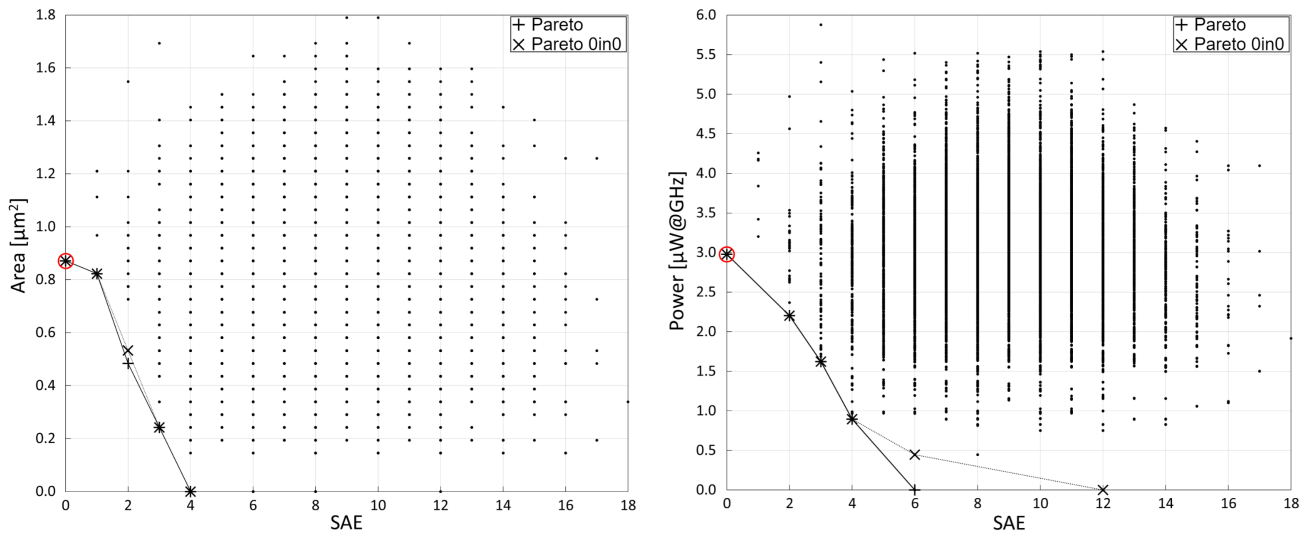


FIGURE 3. Silicon area and power dissipation for the design space of approxFAs vs. the Sum of Absolute Errors. Solid line: Pareto optimal curve for the whole set of approxFAs. Dotted line: Pareto optimal curve for the set of approxFAs that provide 0-output when all input bits are 0.

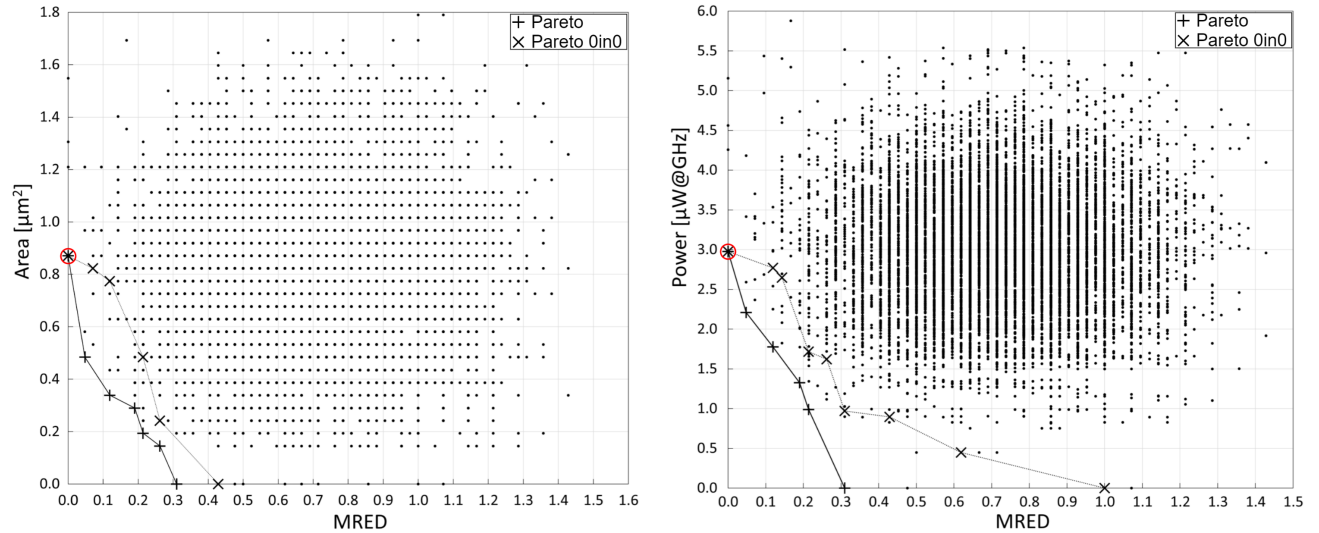


FIGURE 4. Silicon area and power dissipation for the design space of approxFAs vs. Mean Relative Error Distance. Solid line: Pareto optimal curve for the whole set of approxFAs. Dotted line: Pareto optimal curve for the set of approxFAs that provide 0-output when all input bits are 0.

space, numerous FAs are far worse than the exact FA in both area and power dissipation. Notably, as area and power dissipation are highly correlated, architectures inferior to the exact FA in terms of area are very likely to perform worse also in terms of power dissipation. Minimum delay data are not provided since they exhibit similar behavior, showing that many circuits are slower than the exact FA.

Figs. 3 and 4 also report the Pareto fronts designs for the performance factors considered in the figure. These are the designs that provide the best performance in each plot and will be discussed in Section III.C.

B. STATE OF THE ART PROPOSALS

The design space of Figs. 3, 4 includes all the possible designs and therefore also the approxFAs previously proposed in the scientific literature. The placement of these

previously proposed approxFAs in the design space is reported in Figs. 5, 6 where the designs proposed in [9]-[10], [16]-[22], [24], [25], [27], [29]-[37], [39]-[42], [44]-[52] are reported. In many cases the cited papers propose more than one design resulting in a total of 63 circuits taken from the literature. Note that the electrical performance in Figs. 5, 6 refers to the standard cell implementation carried out in this paper and does not include any potential hand-crafted transistor-level or layout-level optimizations considered in previous papers.

Many previously proposed approxFAs coincide since they have the same truth table (or the respective truth tables can be obtained with a permutation of the input operands). A noticeable example is the approxFA whose ID is E817 that has been previously proposed in [17], [21], [22], [24], [29], [30], [35], [47], [48], and [49]. The results in Figs. 5, 6 also

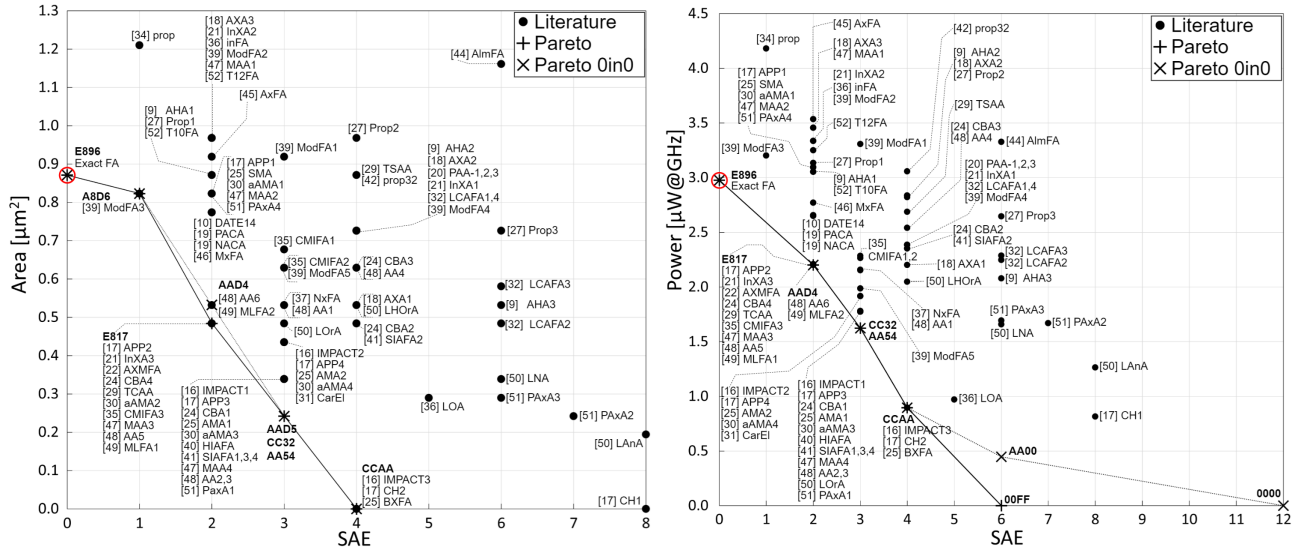


FIGURE 5. Silicon area and power dissipation as a function of the Sum of Absolute Errors. Pareto fronts and the state-of-the-art designs.

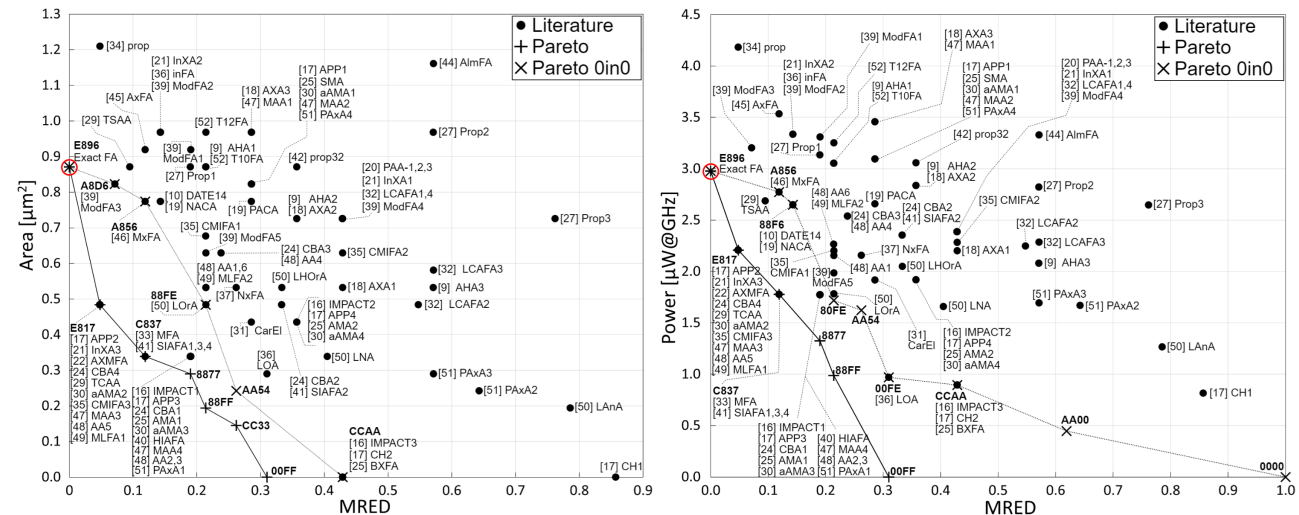


FIGURE 6. Silicon area and power dissipation as a function of the Mean Absolute Relative Error. Pareto fronts and the state-of-the-art designs.

highlight that most of the previously proposed approxFAs are sub-optimal.

C. PARETO FRONTS

The Pareto front circuits constitute the best design choices as they provide the best trade-off for a given couple of electrical-error performance factors.

Figs. 3, 4, 5, and 6 report two Pareto fronts since there are several cases in which there is a requirement for approximate full adders that produce the correct output $\{0,0\}$ when the input is $\{0,0,0\}$. Throughout the paper, this is referred to as the “0-in/0-out property”. If approxFAs lacking this behavior are used, binary adders and multipliers may generate non-zero outputs while performing trivial operations such as the sum or product of two zero operands. This is often unwanted in applications involving approximate multipliers [62].

To investigate the implication of this requirement, we

have extracted the subset of approxFAs having the 0-in/0-out property from the whole set of 13,055 approxFAs. This subset comprises 3,263 circuits, whose performances have been extracted to derive the corresponding Pareto fronts.

Figs. 3, 4, 5, and 6 show the Pareto fronts (given by designs having the best area or power for each error metric), for both the entire set of approxFAs (named Pareto in the figures) and the reduced set of approxFAs having the 0-in/0-out property (named Pareto 0in0 in the figures).

The electrical and error performances of the approxFAs on the Pareto fronts for the whole set of approxFAs are reported in Table IV (Pareto front of the area vs *SAE*), Table V (Pareto front of the power vs *SAE*), Table VI (Pareto front of the area vs *MRED*), and Table VII (Pareto front of the power vs *MRED*).

The electrical and error performances of the approxFAs on the Pareto fronts considering only the subset of designs

TABLE IV
PARETO FRONT APPROXIMATE FA. AREA VS SAE

ID	A8D6	E817	AAD5	CC32	CCAA
SAE	1	2	3	3	4
MRED	0.071	0.048	0.214	0.262	0.429
Error Prob.	0.125	0.25	0.375	0.375	0.5
Mean Error	0.125	0	-0.125	0.125	0
Max Abs. Err.	1	1	1	1	1
Area	94%	56%	28%	28%	0%
Power	108%	74%	57%	55%	30%
Max delay	102ps	68ps	40ps	38ps	6ps

TABLE V
PARETO FRONT APPROXIMATE FA. POWER VS SAE

ID	AAD4	CC32	CCAA	00FF
SAE	2	3	4	6
MRED	0.214	0.262	0.429	0.310
Error Prob.	0.25	0.375	0.5	0.625
Mean Error	0	0.125	0	0.5
Max Abs. Err.	1	1	1	2
Area	61%	28%	0%	0%
Power	74%	55%	30%	0%
Max delay	61ps	38ps	6ps	0ps

TABLE VI
PARETO FRONT APPROXIMATE FA. AREA VS MRED

ID	E817	C837	8877	88FF	CC33	00FF
SAE	2	3	4	4	4	6
MRED	0.048	0.119	0.190	0.214	0.262	0.310
Error Prob.	0.25	0.375	0.5	0.5	0.5	0.625
Mean Error	0	0.125	0.25	0	0	0.5
Max Abs. Err.	1	1	1	1	1	2
Area	56%	39%	33%	22%	17%	0%
Power	74%	60%	45%	33%	38%	0%
Max delay	68ps	75ps	69ps	34ps	28ps	0ps

TABLE VII
PARETO FRONT APPROXIMATE FA. POWER VS MRED

ID	E817	C837	8877	88FF	00FF
SAE	2	3	4	4	6
MRED	0.048	0.119	0.190	0.214	0.310
Error Prob.	0.25	0.375	0.5	0.5	0.625
Mean Error	0	0.125	0.25	0	0.5
Max Abs. Err.	1	1	1	1	2
Area	56%	39%	33%	22%	0%
Power	74%	60%	45%	33%	0%
Max delay	68ps	75ps	69ps	34ps	0ps

having the 0-in/0-out property are listed in Table VIII (area vs *SAE*), Table IX (power vs *SAE*), Table X (area vs *MRED*), and Table XI (power vs *MRED*).

In Tables IV-XI, area and power are reported as a percentage of the value of the exact full adder. Several error metrics, *SAE*, *MRED*, error probability, mean error, and maximum absolute error are reported to clearly describe the error performance of the selected approxFA. The approxFAs highlighted in red correspond to circuits previously presented in the literature. They are: A8D6 (proposed in [39]), E817 (proposed in [17], [21], [22], [24], [29], [30], [35], [47], [48], [49]), CCAA (proposed in [16], [17], [25]), C837 (proposed in [33], [41]), AAD4 (proposed in [48],

[49]), A856 (proposed in [46]), 88FE (proposed in [50]), 88F6 (proposed in [10], [19]), and 00FE (proposed in [36]). The approxFAs 00FF and 0000 correspond to trivial designs with $c_o=0$ and $s=1$, and the truncation with $c_o=0$ and $s=0$, respectively. Note that CCAA is also a trivial design, with $c_o=a$, $s=b$. The other circuits have been discovered by our exhaustive search; among these AA00 is also a trivial design, with $c_o=b$, $s=0$. It is worth highlighting that even if designs, like CCAA, do not use logic, they still have a non-zero delay and power dissipation due to the switching of the output load.

Our analysis in Tables IV-XI shows that some approxFAs are in more than one Pareto fronts. Considering the area and power as electrical performance and *SAE* and *MRED* as error

TABLE VIII
PARETO FRONT APPROXIMATE FA WITH OUTPUT EQUAL TO {0,0}
FOR INPUT {0,0,0}. AREA VS SAE

ID	A8D6	AAD4	AA54	CCAA
SAE	1	2	3	4
MRED	0.071	0.214	0.262	0.429
Error Prob.	0.125	0.25	0.375	0.5
Mean Error	0.125	0	0.125	0
Max Abs. Err.	1	1	1	1
Area	94%	61%	28%	0%
Power	108%	74%	56%	30%
Max delay	102ps	61ps	38ps	6ps

TABLE X
PARETO FRONT APPROXIMATE FA WITH OUTPUT EQUAL TO {0,0} FOR INPUT
{0,0,0}. AREA VS MRED

ID	A8D6	A856	88FE	AA54	CCAA
SAE	1	2	3	3	4
MRED	0.071	0.119	0.214	0.262	0.429
Error Prob.	0.125	0.25	0.375	0.375	0.5
Mean Error	0.125	0.25	0.125	0.125	0
Max Abs. Err.	1	1	1	1	1
Area	94%	89%	56%	28%	0%
Power	108%	93%	60%	56%	30%
Max delay	102ps	78ps	46ps	38ps	6ps

TABLE IX
PARETO FRONT APPROXIMATE FA WITH OUTPUT EQUAL TO {0,0}
FOR INPUT {0,0,0}. POWER VS SAE

ID	AAD4	AA54	CCAA	AA00	0000
SAE	2	3	4	6	12
MRED	0.214	0.262	0.429	0.619	1.0
Error Prob.	0.25	0.375	0.5	0.625	0.875
Mean Error	0	0.125	0	0.5	1.5
Max Abs. Err.	1	1	1	2	3
Area	61%	28%	0%	0%	0%
Power	74%	56%	30%	15%	0%
Max delay	61ps	38ps	6ps	6ps	0ps

TABLE XI
PARETO FRONT APPROXIMATE FA WITH OUTPUT EQUAL TO {0,0} FOR INPUT
{0,0,0}. POWER VS MRED

ID	A856	88F6	80FE	AA54	00FE	CCAA	AA00	0000
SAE	2	2	3	3	5	4	6	12
MRED	0.119	0.143	0.214	0.262	0.310	0.429	0.619	1.0
Error Prob.	0.25	0.25	0.375	0.375	0.5	0.5	0.625	0.875
Mean Error	0.25	0.25	0.375	0.125	0.625	0	0.5	1.5
Max Abs. Err.	1	1	1	1	2	1	2	3
Area	89%	89%	61%	28%	33%	0%	0%	0%
Power	93%	89%	58%	56%	33%	30%	15%	0%
Max delay	78ps	77ps	47ps	38ps	44ps	6ps	6ps	0ps

TABLE XII
SELECTED PARETO FRONT APPROXIMATE FA: ELECTRICAL AND ERROR PERFORMANCE

ID	E817	C837	AAD5	CC32	8877	88FF	CCAA	00FF
SAE	2	3	3	3	4	4	4	6
MRED	0.048	0.119	0.214	0.262	0.190	0.214	0.429	0.310
Error Prob.	0.25	0.375	0.375	0.375	0.5	0.5	0.5	0.625
Mean Error	0	0.125	-0.125	0.125	0.25	0	0	0.5
Max Abs. Err.	1	1	1	1	1	1	1	2
Area	56%	39%	28%	28%	33%	22%	0%	0%
Power	74%	60%	57%	55%	45%	33%	30%	0%
Max delay	68ps	75ps	40ps	38ps	69ps	34ps	6ps	0ps

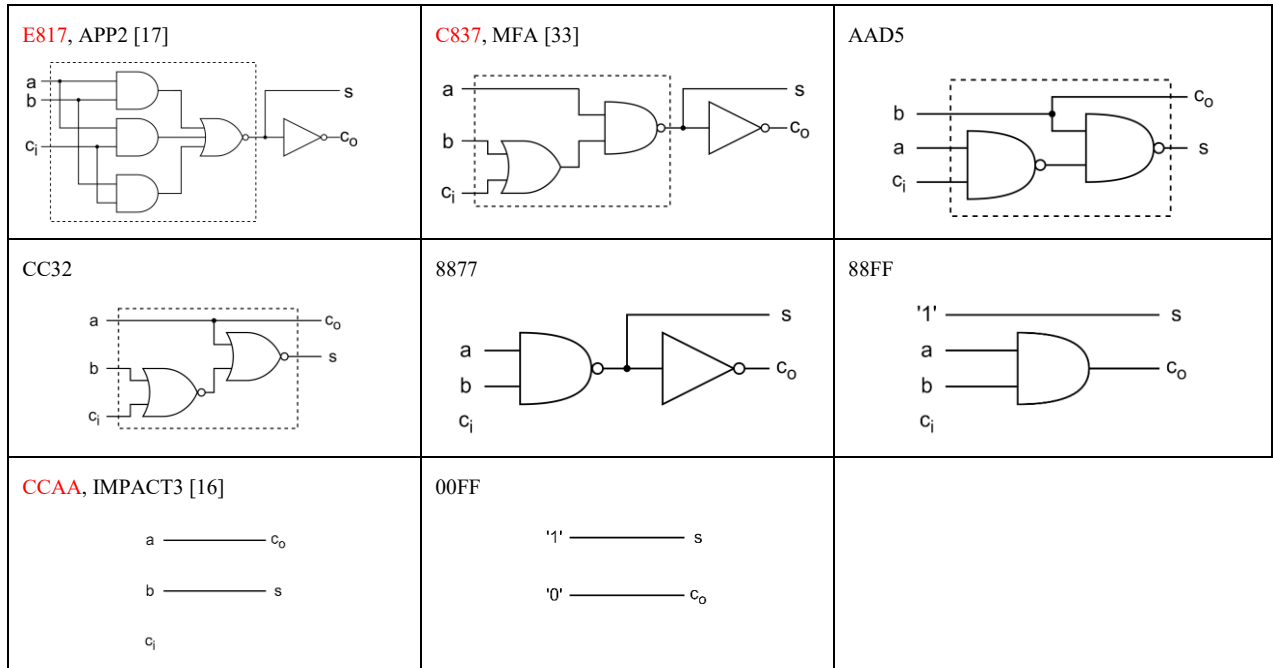


FIGURE 7. Circuit implementation of the approximate FA that are on the Pareto front and are selected due to optimal overall performance.

metrics, we have extracted two sets of suggested approxFAs. The first one considers the entire design space, while the second one considers only the subset of approxFAs having the 0-in/0-out property. The suggested sets include eight and nine approxFAs, respectively. All the other designs are sub-optimal.

D. RECOMMENDED APPROXIMATE FULL ADDERS FROM THE WHOLE SET OF DESIGNS

The recommended approxFAs considering the entire design space are reported in Table XII, while their logic implementation in the considered standard cell library is shown in Fig. 7. The set is composed of eight approxFA designs. Three of them are highlighted in red since they correspond to circuits previously presented in literature. The last two columns in Table XII correspond to designs that use no logic. The design with ID=CCAA corresponds to $s=b$, $c_o=a$, while the design with ID=00FF corresponds to $s=1$, $c_o=0$ with the output stuck to fixed values.

The selection of the proposed set of optimal approximate full adders has been based on the results presented in

Tables IV-VII and on the following observations:

- 1) The design A8D6 provides a 6% area advantage with respect to the exact design but exhibits an 8% power disadvantage and a 38% increased delay when compared with the exact full adder. The possible practical advantages of this design are very limited, and it has not been selected.
- 2) The design E817 has been selected as it is on most Pareto fronts and is a good design choice. As said before, the design has been firstly proposed in [17] as APP2 and re-proposed in the following years with different names. The only design that overcomes APP2 is AAD4 that has hardly less power dissipation and is faster but is far worse in terms of area and MRED.
- 3) For $SAE=3$ the four Pareto fronts suggest three alternatives, C837, AAD5, and CC32 that are selected and listed with increasing MRED and decreasing power dissipation.
- 4) For $SAE=4$ the optimal approxFAs with decreasing power and increasing MRED, are 8877, 88FF, and CCAA. The three designs are all included in Table XII.

TABLE XIII
SELECTED PARETO FRONT APPROXIMATE FA: ELECTRICAL AND ERROR PERFORMANCE WITH OUTPUT EQUAL TO {0,0} FOR INPUT {0,0,0}.

ID	A856	88F6	AAD4	80FE	AA54	00FE	CCAA	AA00	0000
SAE	2	2	2	3	3	5	4	6	12
MRED	0.119	0.143	0.214	0.214	0.262	0.310	0.429	0.619	1.0
Error Prob.	0.25	0.25	0.25	0.375	0.375	0.5	0.5	0.625	0.875
Mean Error	0.25	0.25	0	0.375	0.125	0.625	0	0.5	1.5
Max Abs. Err.	1	1	1	1	1	2	1	2	3
Area	89%	89%	61%	61%	28%	33%	0%	0%	0%
Power	93%	89%	74%	58%	56%	33%	30%	15%	0%
Max delay	78ps	77ps	61ps	47ps	38ps	44ps	6ps	6ps	0ps

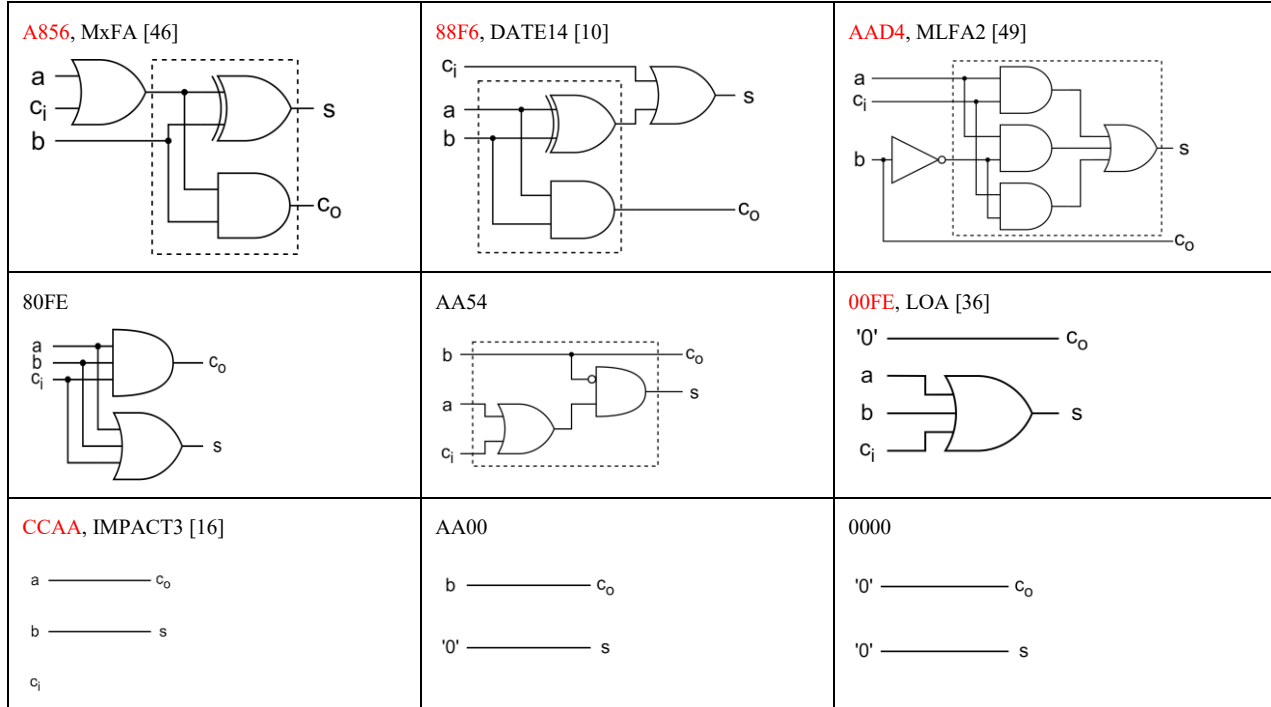


FIGURE 8. Circuit implementation of the Pareto front approxFAs with the 0-in/0-out property, selected due to optimal overall performance.

- 5) For $SAE=5$ no design has been selected since no design belongs to a Pareto optimal front.
- 6) For $SAE=6$ we have the trivial design 00FF ($s=1$, $c_o=0$) that has been included in Table XII.

E. RECOMMENDED APPROXIMATE FULL ADDERS FROM THE SET OF DESIGNS THAT SATISFY THE 0-IN/0-OUT PROPERTY

The selected designs taken from Tables VIII-XI are reported in Table XIII and the logic implementation is in Fig. 8. The set of suggested designs is composed of nine circuits, five of which are highlighted in red indicating that they have been previously proposed in the scientific literature. The last three columns in Table XIII correspond to designs that use no logic. The last column has ID=0000 and corresponds to total truncation since the outputs are stuck to zero. The design with ID=CCAA corresponds to $s=b$, $c_o=a$, while the design with ID=AA00 corresponds to $s=0$, $c_o=b$.

The selection of the proposed set of optimal approximate full adders has been based on the results presented in Tables VIII-XI and on the following observations:

- 1) As also mentioned in Section III.D, the approxFA A8D6 is excluded from the list of recommended designs due to the questionable advantage with respect to the exact design.
- 2) For $SAE=2$, the three designs A856, 88F6, AAD4 (mentioned in decreasing power and increasing MRED order) are included in the list.
- 3) For $SAE=3$, we include the designs 80FE and AA54 (with AA54 appearing on all four Pareto fronts). The design 88FE is excluded from the list since 80FE (with the same SAE and MRED) is slightly more efficient in terms of power.
- 4) For $SAE=4$, $SAE=5$, and $SAE=6$, CCAA, 00FE, and AA00, respectively, are the unique options and are included in the list.
- 5) For $7 \leq SAE \leq 11$, there are no designs.
- 6) For $SAE=12$, we have the trivial design 0000 that is included in the list.

IV. TESTING THE APPROXIMATE FULL-ADDERS IN ERROR RESILIENT APPLICATIONS

In order to conduct a thorough examination of the proposed approximate Full-Adders, three common error-resilient applications have been taken into consideration.

Two image processing applications, image addition and image blurring, and a hardware implementation of an Artificial Neural Network have been examined to confirm that the proposed sets of optimal designs provide good performances in a variety of cases.

In the neural network application, only binary multipliers have been approximated. In the image addition, only carry ripple adders have been approximated. Finally, in the image blurring application, a multi-operand adder and the binary multipliers have both been approximated. In this way the proposed approxFA designs are tested not only in different applications but are also used for the implementation of different types of arithmetic circuits.

The fact that each application focuses on the use of a single type of approximated arithmetic unit provides useful information on the electrical performance and the quality level that the proposed approxFAs can provide in different scenarios. This allows an easier selection of the best approximate full adder for a given design.

It should be noted that more efficient binary adders and multipliers than those used in the investigated applications may be devised; potentially without even the use of approximate full-adder circuits. However, the purpose of these applications is to demonstrate the merit of the investigated circuits, rather than achieve optimal results for the specific applications.

For these applications the previously proposed approxFAs

listed in the references and reported in Figs. 5 and 6, along with the newly discovered ones reported in Tables XII and XIII have been tested, resulting in a total of 72 different evaluated circuits. All investigated circuits have been synthesized in the same 14 nm FinFET technology, using Cadence Genus. The power dissipation has been computed by simulating the final netlist with random input vectors, to effectively account for the switching activity of each node. Area and power values have been obtained synthesizing all circuits with a timing constraint of 1 ns for the image blurring and image addition applications and 10 ns for the NN. Power, area and precision results for all three applications are detailed in Table XIV for the proposed sets of approxFAs reported in Tables XII and XIII. Table XIV also reports the performance of a selected set of designs taken from the literature that perform reasonably well in the selected applications. As shown in the following, the circuits presented in tables XII and XIII tend to perform very well in all applications even if they are not always optimal.

A. NEURAL NETWORK

The inherent probabilistic characteristics of artificial neural networks and their extensive reliance on multiplication operations, make ANNs highly suitable for accommodating approximate multiplying circuits obtaining substantial reduction in power dissipation. This is particularly relevant in the field of smart edge computing where the power budget constraints are a limiting factor. In this field the implemented neural networks are commonly very small to comply with the reduced amount of memory and limited processing power of the CPU.

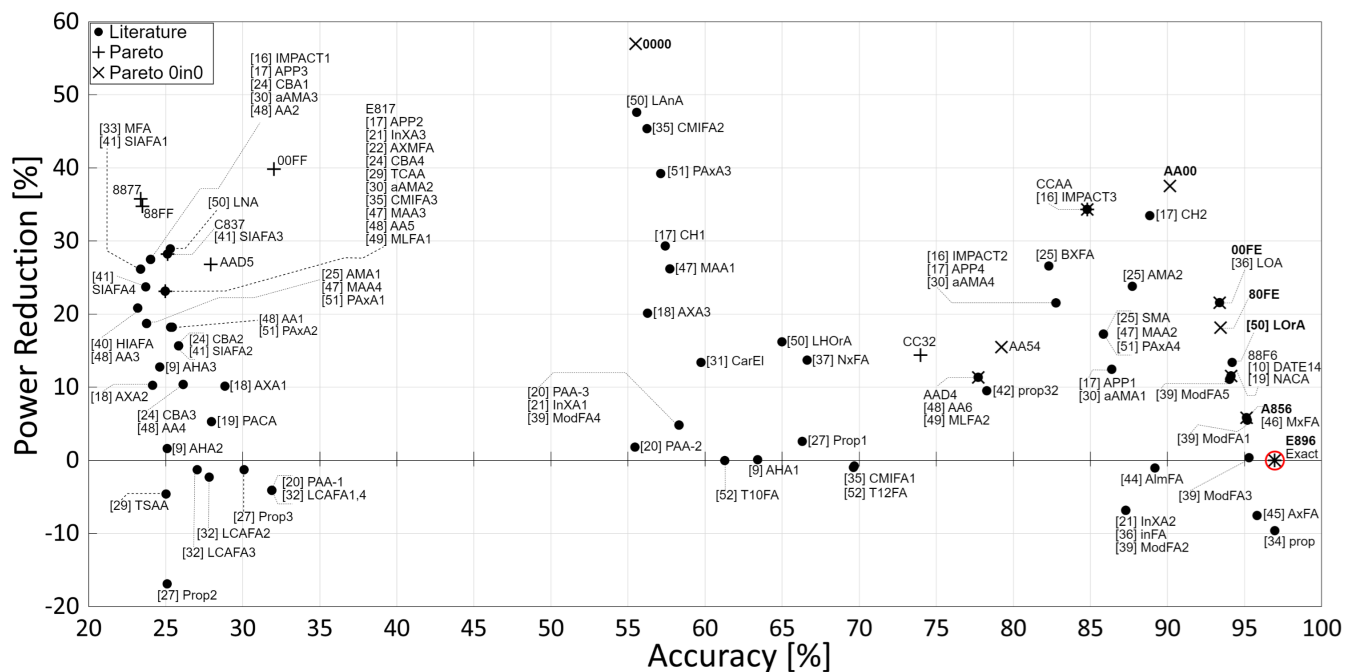


FIGURE 9. Power reduction vs accuracy for a Neural Network using multipliers with different approxFAs. Optimal designs points towards the top right corner exhibiting high accuracy and high power reduction percentage.

Ref. [3], where a detailed overview of the system is provided, proposes a small neural network optimized for edge applications to be implemented in a wearable medical device, and exploits approximate computing to reduce the power dissipation with minimal effect on the accuracy. In [3] machine learning approaches are used to detect and characterize the spiking activity in simulated recordings of brain activity. If detected, a spike is classified as Type-A, B, or C to associate it with the corresponding firing neurons that are randomly placed in a cube of brain tissue. The performance of the neural network is tested modifying both the number of bits of the signal and implementing approximate multipliers. As expected, the longer the selected signal bit-width, the more accurate are the obtained results. In addition, the use of approximate components allows significant power saving with a minimal impact on the accuracy. The HDL description of the network and dataset can be found on GitHub [62].

The neural network circuit considered in this paper has signals represented on 12-bits. The neural network has an input layer of 32 neurons, two hidden layers of 24 and 16 neurons, and an output layer of 4 neurons. The circuit exploits 44 Dadda binary multipliers that are repeatedly activated to compute the more than 1200 multiplications required for each inference.

The approxFAs are exploited in the 44 multipliers used in the inference stage. Specifically, the Dadda structure of the binary multipliers, has been modified so that the exact FAs in the lower half of the partial products reduction matrix, composed by twelve columns, are substituted by the investigated approxFAs. It should be noted, that in the lower

half of the reduction matrix of the 12-bit Dadda multiplier, there are three, twelve, thirteen, eight, and nine full adders in the five reduction stages, respectively. The considered application is a challenging test for the approximate full adders and can demonstrate the effectiveness of the approxFAs considered in this paper. Being a small neural network designed for a wearable medical device, it is a perfect example of edge applications, where power dissipation is a critical concern.

The implementation results are shown in Fig. 9 where the power reduction percentage vs the Accuracy of the neural network is reported. In Fig. 9 the Pareto optimal designs are shown in bold. The results of the proposed sets of designs together with an additional design whose performance is noticeable in this application are reported in Table XIV where accuracy, power reduction percentage, and area reduction percentage are reported. The Neural Network circuits employing the proposed approxFAs that satisfy the suggested 0-in/0-out property show exceptional performance, surpassing almost all the other circuits in this specific application. Namely, A856, 80FE, 00FE, and AA00 from Table XIII, manage to reduce the power consumption from 6% to 38% and reduce the area occupation from 6% to 40% while keeping accuracy rates over 90%, whilst the exact circuit reaches 97% accuracy. Among the circuits that are proposed in the literature and are not included in the Pareto fronts, the approxFA LOrA, proposed in [50] is one of the highest performing circuits for this application and satisfies the 0-in/0-out property. ModFA1 and ModFA5 [39] perform very well in the high-accuracy/low-power-reduction domain but do not outperform the suggested circuits. CC32 and

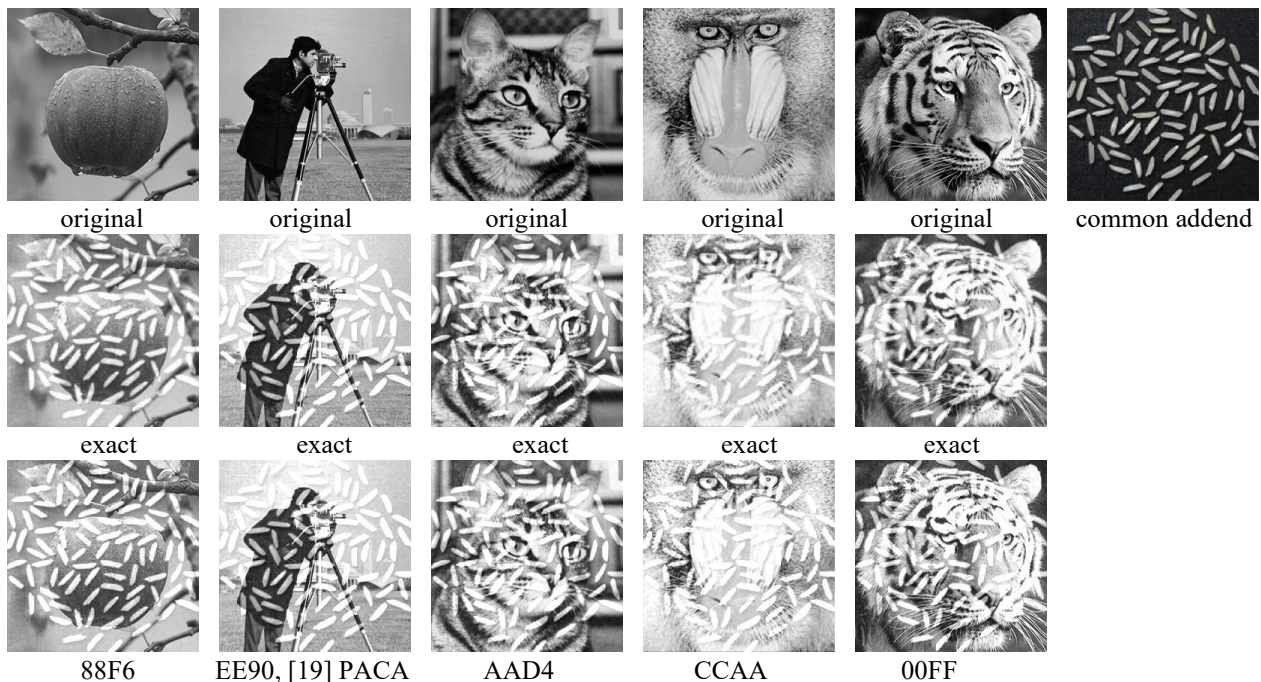


FIGURE 10. Image addition. Five images have been added to the rice image. Images obtained using the exact adder are in the second row. Images obtained from selected approximate adders are in the third row.

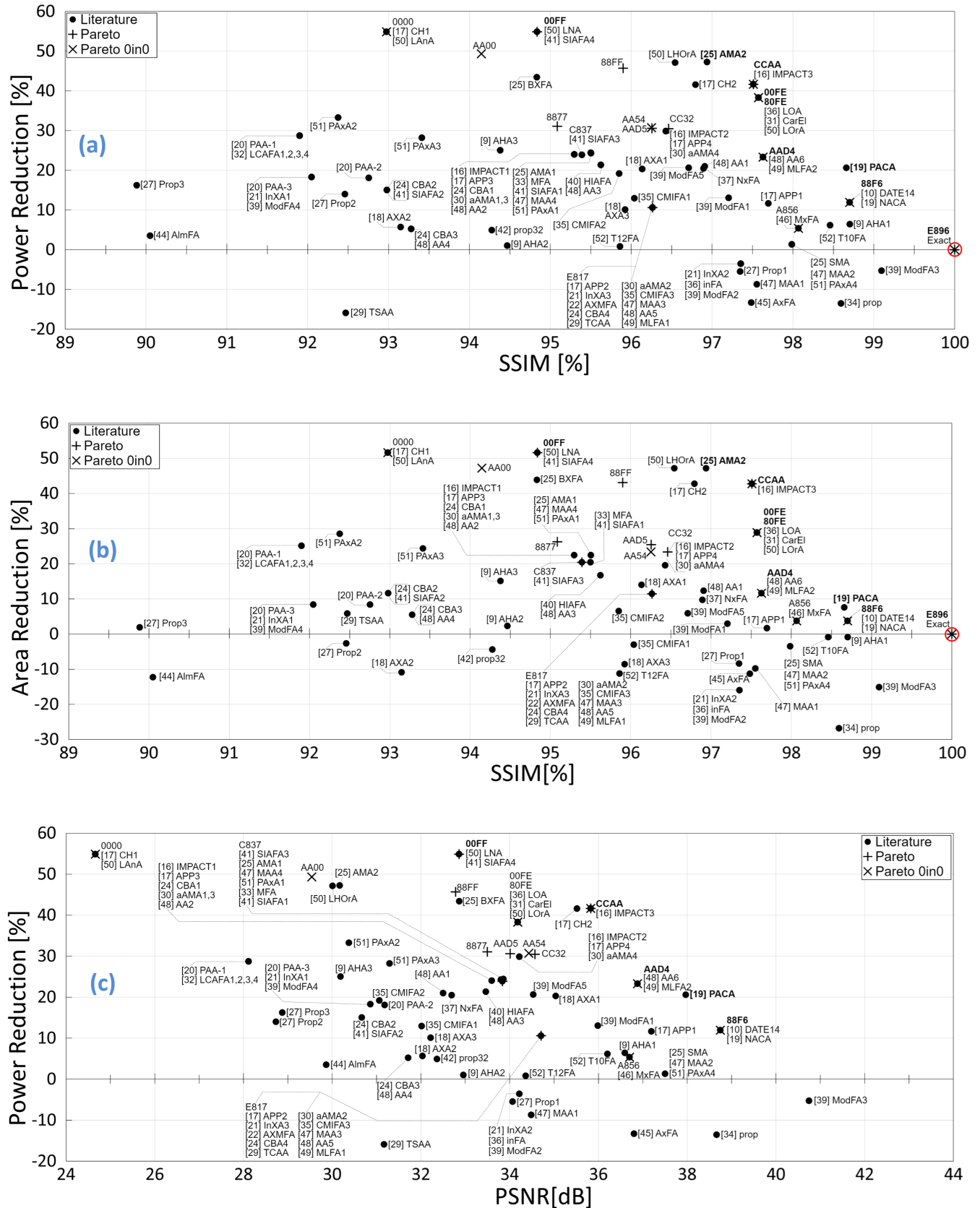


FIGURE 11. Circuits performing image addition using different approxFAs. (a) Power Reduction vs Mean SSIM. (b) Area Reduction vs Mean SSIM. (c) Power Reduction vs Mean PSNR.

CCAA are the only circuits from Table XII that coincidentally have the 0-in/0-out property, perform reasonably well, and they are the most accurate circuits from this set with 74% and 85% accuracy. Their power reduction reaches 14% and 34%, and the area reduction reaches 16% and 33%, respectively.

It should be noted that various versions of the neural network circuit give a mere 25% accuracy, which statistically represents the minimum possible performance for a 4-class problem (three types of spikes and spike absence). These accuracy rates suggest randomness and an obvious lack of reliability.

B. IMAGE ADDITION

Image addition is a common image processing task that combines two images, resulting in a new composite image. The reader can find similar approximate computing applications together with a detailed description of the system in [9], [21], [27], [40], [41]. Each pixel in the produced image corresponds to the mean value of the respective pixels in the original images. A similar application, often used for motion detection is image subtraction [34], [35]. In both cases, Ripple Carry Adders are employed to add the corresponding pixel values, making them suitable for testing approximate full adders.

In this application five grayscale images of 256×256 pixels have been added to a reference image of rice. The color tone is encoded in 8 bits, which is why 8-bit RCAs have been employed to process these signals. The four least significant bits of the signals are added by approxFAs while the rest of the full adders in the RCA chain are exact. The structural similarity index measure (*SSIM*) and the Peak Signal to Noise Ratio (*PSNR*) have been used to measure the similarity between an image composed by an approximate RCA and an image generated by an exact RCA. The images generated by exact RCAs are in the second row of Fig. 10 while images obtained from selected approximate adders are in the third row.

Each circuit in the literature and in the sets of proposed approxFAs has been used in an RCA to add the five images to the image of rice. Mean *SSIM* and mean *PSNR* are used to show the mean performance of each RCA in these five cases.

The results are shown in Fig. 11(a), 11(b), and 11(c) where power reduction vs *SSIM*, area reduction vs *SSIM* and power reduction vs *PSNR* are reported. In Fig. 11 the Pareto optimal results are highlighted in bold. The relevant data are also reported in Table XIV.

In Fig. 11(a), apart from the exact RCA that constitutes the point of reference, there are eight circuits that outperform all the rest. From the set of proposed approxFAs found in Table XIII, 88F6, AAD4, 80FE, 00FE, and CCAA make up the most accurate adders, with mean *SSIMs* from 98.7% to 97.5%. CCAA is also present in the more generic set of proposed approxFAs presented in Table XII, and along with 00FF, they are two of the best performing adders for this

application. Finally, two circuits taken from the literature complete the set of optimal circuits in image addition. AMA2 [25] with a mere 3.1% degradation in mean *SSIM* and a significant 47% reduction in power dissipation and PACA [19] with 1.4% quality degradation and 20.6% power reduction. It is worth highlighting that the reduction in power dissipation is achieved without significantly degrading image quality, as demonstrated in the third row of Fig. 10.

Similar results are obtained from the analysis of Figs. 11(b)-(c). The similarity in Fig. 11(b) is mostly attributed to the high correlation between power dissipation and area occupation while the results of Fig 11(c), which are based on a different performance metric, validate the proposed results.

C. IMAGE BLURRING

A low pass filter in image processing results in image smoothing (blurring) by averaging neighboring pixel values, thus minimizing rapid change between them and eliminating the high spatial frequency noise. The most straightforward low-pass filter consists in replacing each image pixel with the average value of that pixel and its surrounding eight pixels. For a more refined smoothing, the use of a two dimensional, rotationally symmetric, 3×3 Gaussian kernel requires calculating the weighted average value of the nine pixels according to the values of the kernel.

For each pixel in the image, the aforementioned process involves nine multiplications between pixels and kernel values, and the addition of these nine products. Moreover, image blurring is an error resilient application, given that the human eye cannot discern trivial details in blurred images. For these reasons, image smoothing is often used to evaluate the performance of approximate blocks. The reader can find similar applications together with a detailed description of the system in [22], [38], [61].

In this work, a combination of exact and approximate full adders has been used in the partial product reduction stage of the Dadda multipliers, as well as in the adders. Specifically, the image blurring circuit is composed of nine binary multipliers and a final multi-operand adder. ApproxFAs are exploited in the nine and eleven least significant columns of the binary multipliers and the adder, respectively. The Gaussian filter used in this application has a standard deviation equal to 0.9. The floating-point numbers of the kernel have been multiplied by 2^{10} and then rounded, to be in range for the considered 8-bit input multipliers. The matlab code that provides the kernel is the following:

```
round(2^10*fspecial('gaussian',[3 3],0.9))
```

Test images and their blurred counterparts obtained using non-approximate circuits are shown in first and second rows of Fig. 12, respectively. Blurred images obtained by using selected approximate circuits are shown in the third row of Fig. 12. The *SSIM* and the *PSNR* of each blurred image are calculated with respect to the corresponding image blurred

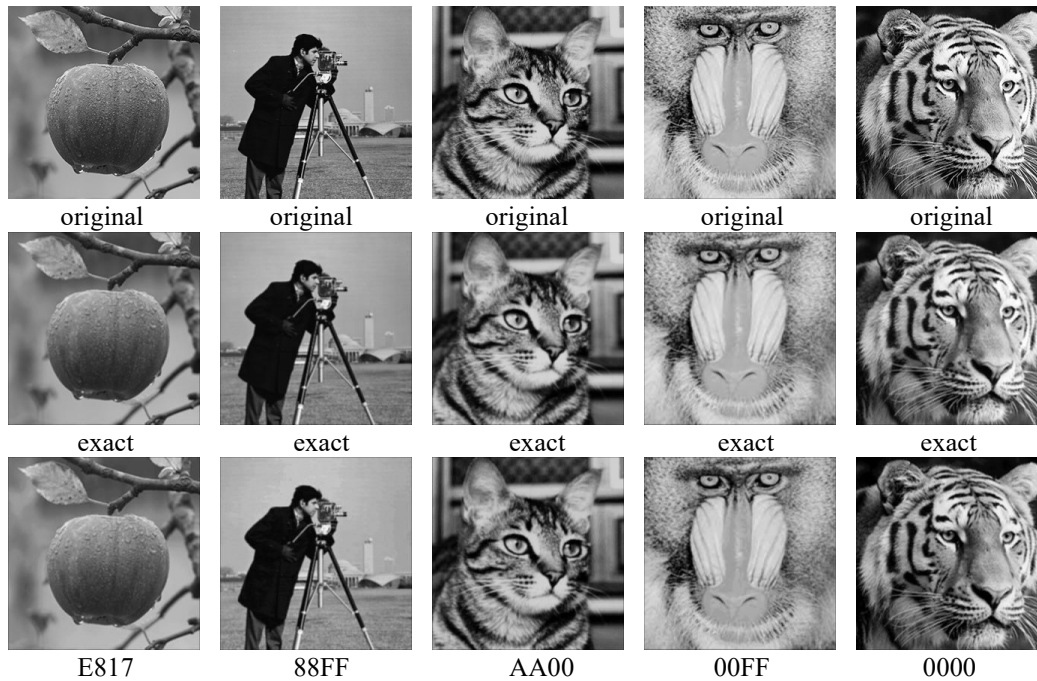


FIGURE 12. Image blurring. Five images have been smoothed using a 3x3 gaussian kernel. Images obtained using the exact smoother are in the second row. Images obtained from selected approximate adders are in the third row.

without approximations. The average *SSIM* and average *PSNR* between all five test images are then calculated to get the performance of each approxFA. Fig. 13(a) shows the trade-off between power reduction and mean *SSIM*. Figs. 13(b)-(c) show the area reduction percentage vs mean *SSIM* trade-off, and the power reduction percentage vs mean *PSNR* trade-off, respectively. In this particular application there are many designs that provide similar performances. These results in Fig. 13 are highlighted with a red circle and zoomed for a detailed view in the provided inset. The asterisk in a red circle represents the exact smoother. In Fig. 13(a) the exact circuit, being the point of reference, offers 100% mean *SSIM* and no power reduction.

Six circuits demonstrate optimal performance in the power vs *SSIM* trade-off in this specific application. From the list of Pareto optimal approxFAs that have the 0-in/0-out property (Table XIII), three relatively low accuracy approxFAs are extracted. 80FE, AA00, and 0000 offer acceptable similarity rates from 98.6% to 95.6% while achieving substantial reductions in power consumption, from 50% to 82%, and in area occupation from 29% to 69%. Three more circuits from the Pareto optimal approxFAs (Table XII) complete the list. E817, 88FF, and 00FF cover the accuracy range from 98.8% to 96.9%, with power reductions from 39.4% to 79.7% and area occupation reduction from 23% to 67%. Prop1 [27] performs quite well, reducing power consumption by 16.5% with a mean *SSIM* of 98.9% but, as shown in Fig.13(a), it is very close to E817 in terms of mean *SSIM*, which reduces power by an additional 23%. Very similar results are obtained when considering the area reduction percentage vs *SSIM* trade-off where E817,

88FF, AA00, 00FF and 0000 are still defining the list of the best performing circuits. In addition, designs A856 from Table XII and SIAFA4 proposed in [41] are among the highest performing circuits for this application. The power reduction percentage vs *PSNR* trade-off highlights a Pareto front composed by AAD4, AA54, LOrA [50], BXFA [25], 88FF, AA00, 00FF, 0000 with four designs out of eight that belong to the suggested designs of Tables XIII (designs with the 0-in/0-out property) and two designs out of eight that are in Table XIV (designs without the 0-in/0-out property).

V. CONCLUSION

In this paper an extensive analysis of approximate full adders has been presented. Like the exact full adder, all the investigated approximate designs encode the sum of three bits with equal weight, into one bit with the same weight and one with double weight. The entire set of such circuits consists of $2^{16}=65536$ designs, including the exact full adder, while if permutations of the same TTs are removed, the final set is reduced to 13055 independent approximate circuits and the exact Full Adder.

The error performance of all the circuits has been derived by performing exhaustive simulations, while the electrical performance has been obtained by synthesizing the investigated circuits in a 14 nm FinFET technology. Results show that a considerable number of approxFAs offer no advantage in power, area, or speed with respect to the exact full adder. Among the remaining approxFAs, two sets of eight and nine suggested approximate full adders have been extracted. One set represents the best among all the possible approxFAs, while the other consists of the best approxFAs

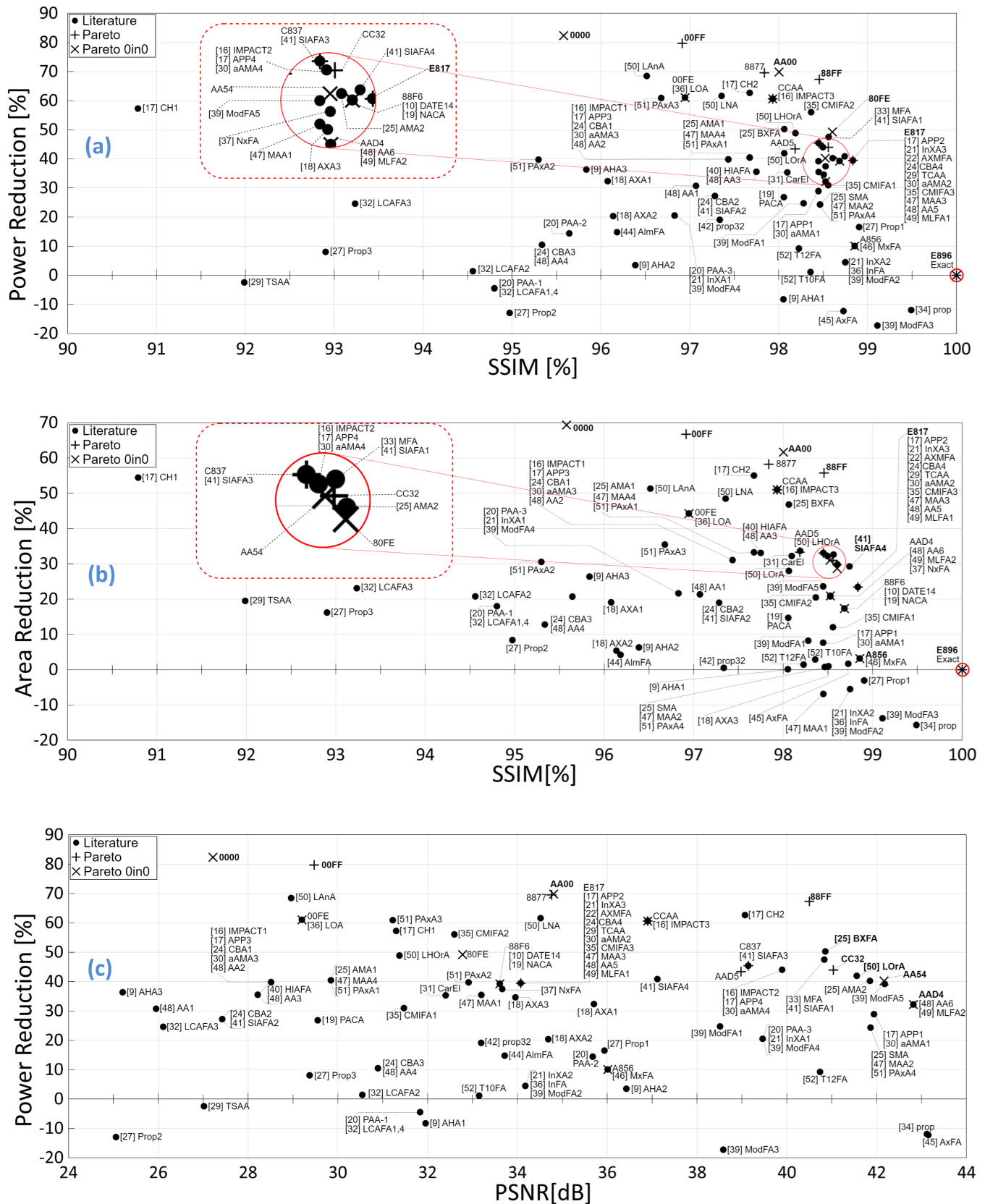


FIGURE 13. Circuits performing image blurring using different approxFAs. (a) Power Reduction vs Mean SSIM. (b) Area Reduction vs Mean SSIM. (c) Power Reduction vs Mean PSNR.

TABLE XIV
PERFORMANCES OF PROPOSED APPROXIMATE FAS IN APPLICATIONS

Design	Neural Network, [3]			Image Addition				Image Blurring				
	Acc.	Power Reduction	Area Reduction	Mean <i>SSIM</i>	Mean <i>PSNR</i>	Power Reduction	Area Reduction	Mean <i>SSIM</i>	Mean <i>PSNR</i>	Power Reduction	Area Reduction	
	(%)	(%)	(%)	(%)	(dB)	(%)	(%)	(%)	(dB)	(%)	(%)	
Suggested designs from Table XII	E817	24.97	23.11	18.35	96.27	34.71	10.62	11.43	98.84	34.07	39.44	23.39
	C837	25.11	28.20	28.85	95.39	33.84	23.90	20.40	98.45	39.14	45.42	33.07
	AAD5	27.92	26.78	30.76	96.26	34.01	30.60	25.45	98.18	38.98	43.41	33.31
	CC32	73.98	14.40	15.77	96.46	34.57	30.46	23.40	98.56	41.04	43.98	30.99
	8877	23.37	35.76	40.55	95.09	33.50	31.09	26.26	97.84	34.76	69.53	58.25
	88FF	23.49	34.75	39.04	95.90	32.78	45.72	43.13	98.46	40.50	67.34	55.76
	CCAA	84.79	34.31	33.48	97.51	35.83	41.66	42.77	97.93	36.90	60.63	50.98
	00FF	32.03	39.83	44.69	94.84	32.86	54.87	51.64	96.92	29.47	79.68	66.75
Suggested designs from Table XIII (0-in/0-out property)	A856	95.13	5.81	6.04	98.07	36.70	5.37	3.77	98.86	36.01	10.02	3.15
	88F6	94.13	11.52	10.65	98.70	38.75	11.94	3.77	98.69	33.61	39.25	17.35
	AAD4	77.72	11.32	11.44	97.63	36.88	23.30	11.60	98.53	42.82	32.21	20.94
	80FE	93.45	18.13	22.76	97.57	34.18	38.32	28.86	98.60	32.78	49.21	28.71
	AA54	79.22	15.47	17.16	96.25	34.43	30.72	23.40	98.52	42.16	40.24	31.02
	00FE	93.38	21.54	27.94	97.57	34.18	38.32	28.86	96.95	29.20	61.01	44.22
	CCAA	84.79	34.31	33.48	97.51	35.83	41.66	42.77	97.93	36.90	60.63	50.98
	AA00	90.16	37.52	39.76	94.15	29.54	49.33	47.24	98.00	34.80	69.81	61.63
	0000	55.49	56.98	57.15	92.97	24.66	54.87	51.64	95.58	27.22	82.37	69.37
LOrA [50]	C0FE	94.20	13.37	16.15	97.57	34.18	38.32	28.86	98.06	41.56	41.92	27.98
SIAFA4 [41]	E01F	23.71	23.71	21.60	94.84	32.86	40.85	29.28	98.74	37.12	40.85	29.28
BXFA [25]	CCF0	82.30	26.56	28.67	94.83	32.87	43.42	43.89	98.06	40.86	50.23	46.76
AMA2 [25]	F08A	87.72	23.79	22.29	96.94	30.17	47.27	47.17	98.61	41.85	40.24	29.84
PACA [19]	EE90	27.98	5.27	8.75	98.66	37.97	20.61	7.59	98.06	29.55	26.78	14.68

that provide a correct output in case all inputs are low.

A Neural Network, a carry ripple adder application, and a combination of an 8-bit multiplier and a nine-inputs adder in an image blurring application have been developed to test the merit of the proposed circuits. When it comes to power dissipation and area occupation, the circuits that belong in the suggested sets of approxFAs (reported in Tables XII and XIII) perform very well, always offering a reduction, a characteristic not always true for the designs reported in the scientific literature. In terms of accuracy, even if the designs do not always provide the best trade-off for a given range, they still have good or optimal performances consistently. These results confirm that the selected approximate designs exhibit a reliable behavior and can be profitably used to alleviate the hardware burden in digital arithmetic circuits. Therefore, designers seeking an approximate component can conveniently resort to one of our limited sets of suggested solutions, avoiding the need to delve into the huge list of potential approximate full adders.

REFERENCES

- [1] J. N. Mitchell, "Computer Multiplication and Division Using Binary Logarithms," in *IRE Transactions on Electronic Computers*, vol. EC-11, no. 4, pp. 512-517, Aug. 1962, 10.1109/TEC.1962.5219391.
- [2] V. K. Chippa, S. T. Chakradhar, K. Roy and A. Raghunathan, "Analysis and characterization of inherent application resilience for approximate computing," in *Proc. 50th ACM/EDAC/IEEE Design Automation Conf. (DAC)*, Austin, TX, 2013, pp. 1-9.
- [3] E. Zacharelos, C. Scognamiglio, E. Napoli and D. Gragnaniello, "On-chip spike detection and classification using neural networks and approximate computing," in *2023 Artificial Intelligence BioMedical Circuits And Systems For Health (BioCAS)*, Toronto, Canada, 2023, pp. 1-5, 10.1109/BioCAS58349.2023.10388854.
- [4] M. Breuer, "Hardware that produces bounded rather than exact results," in *Design Automation Conference*, Anaheim, CA, USA, 2010, pp. 871-876, 10.1145/1837274.1837493.
- [5] R. Venkatesan, A. Agarwal, K. Roy and A. Raghunathan, "MACACO: Modeling and analysis of circuits for approximate computing," in *2011 IEEE/ACM Int. Conf. on Computer-Aided Design (ICCAD)*, San Jose, CA, USA, 2011, pp.667-673, 10.1109/ICCAD.2011.6105401.
- [6] J. Han and M. Orshansky, "Approximate computing: an emerging paradigm for energy-efficient design," in *proc. 18th IEEE Europ. Test Symp.*, Avignon, France, May 2013, pp. 1-6, 10.1109/ETS.2013.6569370.
- [7] M. Horowitz, "1.1 Computing's energy problem (and what we can do about it)," in *Proc. 2014 IEEE Int. Solid-State Circuits Conf. (ISSCC)*, San Francisco, CA, 2014, pp. 10-14, 10.1109/ISSCC.2014.6757323.
- [8] Q. Xu, T. Mytkowicz and N. S. Kim, "Approximate Computing: A Survey," in *IEEE Design & Test*, vol. 33, no. 1, pp. 8-22, Feb. 2016.
- [9] C. K. Jha, S. N. Ved, I. Anand and J. Meki, "Energy and Error Analysis Framework for Approximate Computing in Mobile Applications," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 67, no. 2, pp. 385-389, Feb. 2020, 10.1109/TCSII.2019.2910137.
- [10] C. Liu, J. Han and F. Lombardi, "A low-power, high-performance approximate multiplier with configurable partial error recovery," in *2014 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Dresden, Germany, 2014, pp. 1-4, 10.7873/DATE.2014.108.
- [11] A. Cilaro, et al., "High Speed Speculative Multipliers Based on Speculative Carry-Save Tree," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 61, no. 12, pp. 3426-3435, Dec. 2014, 10.1109/TCSI.2014.2337231.
- [12] J. Melchert, S. Behroozi, J. Li and Y. Kim, "SAADI-EC: A Quality-Configurable Approximate Divider for Energy Efficiency," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 27, no. 11, pp. 2680-2692, Nov. 2019, 10.1109/TVLSI.2019.2926083.

- [13] D. De Caro, N. Petra, A. G. M. Strollo, F. Tessitore and E. Napoli, "Fixed-Width Multipliers and Multipliers-Accumulators With Min-Max Approximation Error," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 60, no. 9, pp. 2375-2388, Sept. 2013, 10.1109/TCSI.2013.2245252.
- [14] Ying Wang, Huawei Li and Xiaowei Li, "Real-time meets approximate computing: An elastic CNN inference accelerator with adaptive trade-off between QoS and QoR," in *2017 54th ACM/EDAC/IEEE Design Automation Conference (DAC)*, Austin, TX, 2017, pp. 1-6, 10.1145/3061639.3062307.
- [15] B. J. Phillips, D. R. Kelly, and B. W. Ng, "Estimating adders for a low-density parity check decoder," in *Proceedings of SPIE*, vol. 6313, 2006, pp. 1-9, 10.1117/12.680199.
- [16] V. Gupta, D. Mohapatra, S.P. Park, A. Raghunathan and K. Roy, "IMPACT: IMPrecise adders for low-power approximate computing," in *IEEE/ACM Int. Symp. on Low Power Electronics and Design*, Fukuoka, Japan, 2011, pp. 409-414, 10.1109/ISLPED.2011.5993675.
- [17] V. Gupta, D. Mohapatra, A. Raghunathan and K. Roy, "Low-Power Digital Signal Processing Using Approximate Adders," in *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 32, no. 1, pp. 124-137, Jan. 2013, 10.1109/TCAD.2012.2217962.
- [18] Z. Yang, A. Jain, J. Liang, J. Han and F. Lombardi, "Approximate XOR/XNOR-based adders for inexact computing," in *2013 13th IEEE Int. Conf. on Nanotechnology (IEEE-NANO 2013)*, Beijing, China, 2013, pp. 690-693, 10.1109/NANO.2013.6720793.
- [19] X. Fan, T. Zhang, H. Li, H. Liu, S. Lu and J. Han, "DACA: Dynamic Accuracy-Configurable Adders for Energy-Efficient Multi-Precision Computing," in *IEEE Transactions on Nanotechnology*, vol. 22, pp. 400-408, 2023, 10.1109/TNANO.2023.3297325.
- [20] M. V. K. Reddy, N. R. S. Reddy, P. N. Kumar, R. P. K. Reddy, M. Sarada and A. Srinivasulu, "Modified Low Power High Speed Approximate Adders For Energy Efficient Arithmetic Applications," in *2023 15th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, Bucharest, Romania, 2023, pp. 01-04, 10.1109/ECAI58194.2023.10193956.
- [21] H. A. F. Almurib, T. N. Kumar and F. Lombardi, "Inexact designs for approximate low power addition by cell replacement," in *2016 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Dresden, Germany, 2016, pp. 660-665.
- [22] G. Gulafshan, R. Kumar, D. Khan, S. Amara and Y. Massoud, "Power Efficient Image Processing with TMR Tunable Hybrid Approximate Adders," in *2023 IEEE 23rd International Conference on Nanotechnology (NANO)*, Jeju City, Korea, 2023, pp. 556-561, 10.1109/NANO58406.2023.10231223.
- [23] M. Shafique, R. Hafiz, S. Rehman, W. El-Harouni and J. Henkel, "Invited: Cross-layer approximate computing: From logic to architectures," in *2016 53rd ACM/EDAC/IEEE Design Automation Conference (DAC)*, Austin, TX, USA, 2016, pp. 1-6, 10.1145/2897937.2905008.
- [24] M. Ramasamy, G. Narmadha and S. Deivasigamani, "Carry based approximate full adder for low power approximate computing," in *2019 7th Int. Conf. on Smart Computing & Communications (ICSCC)*, Sarawak, Malaysia, 2019, pp. 1-4, 10.1109/ICSCC.2019.8843644.
- [25] P. A. Silva and C. Meinhardt, "Energy-Efficient Design of Approximated Full Adders," in *2020 27th IEEE Int. Conf. on Electronics, Circuits and Systems (ICECS)*, Glasgow, UK, 2020, pp. 1-4, 10.1109/ICECS49266.2020.9294925.
- [26] S. Mohanraj and M. Maheswari, "SERF and modified SERF adders for ultra low power design techniques," in *Procedia Engineering*, vol. 30, pp. 639-645, Dec. 2012, 10.1016/j.proeng.2012.01.909.
- [27] C. V. Gowdar, M. C. Parameshwara and S. Sonoli, "Approximate Full Adders for Multimedia Processing Applications," in *2020 IEEE Int. Conf. for Innovation in Technology (INOCON)*, Bangluru, India, 2020, pp. 1-4, 10.1109/INOCON50539.2020.9298237.
- [28] W. Yang, L. Wang and K. Li, "An Approximate Adder Design Based on Inexact Full Adders," in *2021 IEEE 15th Int. Conf. on Anti-counterfeiting, Security, and Identification (ASID)*, Xiamen, China, 2021, pp. 154-158, 10.1109/ASID52932.2021.9651713.
- [29] W. Yang and H. Thapliyal, "Low-Power and Energy-Efficient Full Adders With Approximate Adiabatic Logic for Edge Computing," in *2020 IEEE Comp. Society Annual Symp. on VLSI (ISVLSI)*, Limassol, Cyprus, 2020, pp. 312-315, 10.1109/ISVLSI49217.2020.00064.
- [30] Y. Shanmugam, S. Chandran, C. Rohith and R. Silambarasan, "Design and Analysis of Approximate Adders Using Multiplexer," in *2021 7th Int. Conf. on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, 2021, pp. 977-980, 10.1109/ICACCS51430.2021.9441853.
- [31] B. Sudheshna, C. Vudadha, P. Wali and G. K. Kamath, "An Approximate Full-Adder to Eliminate Carry Propagation in Lower Significant Stage of a Multi-Digit Adder," in *2023 International Conference on Computer, Electronics & Electrical Engineering & their Applications (IC2E3)*, Srinagar Garhwal, India, 2023, pp. 1-6, 10.1109/IC2E357697.2023.10262637.
- [32] A. Yan et al., "Design of Low-Cost Approximate CMOS Full Adders," in *2023 IEEE International Symposium on Circuits and Systems (ISCAS)*, Monterey, CA, USA, 2023, pp. 1-5, 10.1109/ISCAS46773.2023.10181531.
- [33] A. Rampeesa, P. Akhila, M. Irfan, S. Rebelli, L. R. Thoutam and J. Ajayan, "Design of Low Power 4-Bit Baugh-Wooley Multiplier using 1-Bit Mirror and Approximate Full Adders," in *2022 2nd Asian Conference on Innovation in Technology (ASIANCON)*, Ravet, India, 2022, pp. 1-4, 10.1109/ASIANCON55314.2022.9908919.
- [34] S. E. Fatemeh and M. R. Reshadinezhad, "Power-efficient, high-PSNR approximate full adder applied in error-resilient computations based on CNTFETs," in *2020 20th International Symposium on Computer Architecture and Digital Systems (CADS)*, Rasht, Iran, 2020, pp. 1-5, 10.1109/CADS50570.2020.9211854.
- [35] Z. Zareei, K. Navi and P. Keshavarziyan, "Low-power, high-speed 1-bit inexact Full Adder cell designs applicable to low-energy image processing," in *International Journal of Electronics*, vol. 105, pp. 375-384, Jul. 2017, 10.1080/00207217.2017.1357207.
- [36] Y. Guo, H. Sun and S. Kimura, "Design of Power and Area Efficient Low-Part-OR Approximate Multiplier," in *TENCON 2018 - 2018 IEEE Region 10 Conference*, Jeju, Korea (South), 2018, pp. 2110-2115, 10.1109/TENCON.2018.8650108.
- [37] H. Waris, C. Wang, W. Liu, J. Han and F. Lombardi, "Hybrid Partial Product-Based High-Performance Approximate Recursive Multipliers," in *IEEE Transactions on Emerging Topics in Computing*, vol. 10, no. 1, pp. 507-513, 1 Jan.-March 2022, doi:10.1109/TETC.2020.3013977.
- [38] D. Esposito, A. G. M. Strollo, E. Napoli, D. De Caro and N. Petra, "Approximate Multipliers Based on New Approximate Compressors," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 65, no. 12, pp. 4169-4182, Dec. 2018, 10.1109/TCSI.2018.2839266.
- [39] D. P. A. Krishnamurthy, B. Rakshith, G. Prakruthi and Sushma, "Leveraging Approximate 4:2 Compressors for Improved Performance in an 8-Bit Dadda Multiplier Design," in *2023 1st International Conference on Circuits, Power and Intelligent Systems (CCPIS)*, Bhubaneswar, India, 2023, pp. 1-6, 10.1109/CCPIS59145.2023.10292058.
- [40] F. Seiler and N. TaheriNejad, "An IMPLY-based Semi-Serial Approximate In-Memristor Adder," in *2023 IEEE Nordic Circuits and Systems Conference (NorCAS)*, Aalborg, Denmark, 2023, pp. 1-7, 10.1109/NorCAS58970.2023.10305490.
- [41] S. E. Fatemeh, M. R. Reshadinezhad and N. TaheriNejad, "Fast and Compact Serial IMPLY-Based Approximate Full Adders Applied in Image Processing," in *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 13, no. 1, pp. 175-188, March 2023, 10.1109/JETCAS.2023.3241012.
- [42] S. Venkatachalam and S.-B. Ko, "Design of Power and Area Efficient Approximate Multipliers," in *IEEE Transactions on Very Large Scale Integration Systems*, vol. 25, no. 5, pp. 1782-1786, 2017, 10.1109/TVLSI.2016.2643639.
- [43] Z. Chu, C. Shang, T. Zhang, Y. Xia, L. Wang and W. Liu, "Efficient Design of Majority-Logic-Based Approximate Arithmetic Circuits," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 30, no. 12, pp. 1827-1839, Dec. 2022, 10.1109/TVLSI.2022.3210252.
- [44] K. C. Pathak et al., "An Efficient Dadda Multiplier using Approximate Adder," in *2020 IEEE REGION 10 CONFERENCE (TENCON)*, Japan, 2020, pp. 176-181, 10.1109/TENCON50793.2020.9293737.

- [45] S. Pabithra, and S. Nageswari, "Analysis of Approximate Multiplier Using 15-4 Compressor for Error Tolerant Application," in *2018 International Conference on Control, Power, Communication and Computing Technologies (ICCPCT)*, Kannur, India, 2018, pp. 410-415, 10.1109/ICCPCT.2018.8574287.
- [46] N. Kumar M., R. S. Adithyaa, B. Kumar D., and T. Pavithra, "Design Analysis of Wallace Tree based Multiplier using Approximate Full Adder and Kogge Stone Adder," in *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*, India, 2020, pp. 612-616, 10.1109/ICACCS48705.2020.9074336.
- [47] K. M. Reddy, Y. B. Nithin Kumar, D. Sharma, and M. H. Vasantha, "Low power, high speed error tolerant multiplier using approximate adders," in *2015 19th International Symposium on VLSI Design and Test*, India, 2015, pp. 1-6, 10.1109/ISVDT.2015.7208150.
- [48] R. V. Nithyashree, S. Afreen and S. Tantry, "Analysis of various Approximate adders in Ripple Carry Adder design.," in *2023 4th IEEE Global Conference for Advancement in Technology (GCAT)*, India, 2023, pp. 1-5, 10.1109/GCAT59970.2023.10353293.
- [49] P. Lagidi, A. Iswarya, G. Rajesh, and A. S. Keerthi Nayani, "Design of 16-Bit and 32-Bit Approximate Full Adder Using Majority Logic," in *2021 2nd Global Conference for Advancement in Technology (GCAT)*, India, 2021, pp. 1-5, 10.1109/GCAT52182.2021.9587782.
- [50] D. Amuru, A. Zahra, V. Karakuram, and Z. Abbas, "Design of Approximate Full Adders for Error Resilient Applications," in *2023 International Conference on Computer and Applications (ICCA)*, Cairo, Egypt, 2023, pp. 1-6, 10.1109/ICCA59364.2023.10401821.
- [51] A. Kumar, S. Sinha, and D. Bansal, "Design of Efficient Approximate Adders for Error Analysis and Mitigation," in *2023 International Conference on Ambient Intelligence, Knowledge Informatics and Industrial Electronics (AIKIIIE)*, Ballari, India, 2023, pp. 1-5, 10.1109/AIKIIIE60097.2023.10390198.
- [52] K. Hasan, S. M. Nur Pranto, S. Biswas, F. Rabby, M. A. Rahman, and M. Anwarul Abedin, "Design of Pass Transistor-Based Low-Power Approximate Adders for DSP Application," in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2023, pp. 1-6, 10.1109/ICCIT60459.2023.10441300.
- [53] A. Senobari, J. Vafaei, O. Akbari, C. Hochberger and M. Shafique, "A Quality-Aware Voltage Overscaling Framework to Improve the Energy Efficiency and Lifetime of TPUs Based on Statistical Error Modeling," in *IEEE Access*, vol. 12, pp. 92181-92197, 2024, doi: 10.1109/ACCESS.2024.3422012.
- [54] A. Momeni, J. Han, P. Montuschi and F. Lombardi, "Design and Analysis of Approximate Compressors for Multiplication," in *IEEE Transactions on Computers*, vol. 64, no. 4, pp. 984-994, April 2015, doi: 10.1109/TC.2014.2308214.
- [55] E. Napoli, A. G. M. Strollo, E. Zacharelos and G. Di Meo, "Comprehensive Analysis of Input Order Invariant Approximate 4-2 Compressors for Binary Multipliers," *2024 IEEE International Symposium on Circuits and Systems (ISCAS)*, Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558503.
- [56] M. Ahmadinejad, M. H. Moaiyeri and F. Sabetzadeh, "Energy and Area Efficient Imprecise Compressors for Approximate Multiplication at Nanoscale", *International Journal of Electronics and Communications (AEU)*, vol. 110, pp. 4200-4208, October. 2019
- [57] E. Zacharelos, I. Nunziata, G. Saggese, A. G. M. Strollo and E. Napoli, "Approximate Recursive Multipliers Using Low Power Building Blocks," in *IEEE Transactions on Emerging Topics in Computing*, vol. 10, no. 3, pp. 1315-1330, 1 July-Sept. 2022, doi: 10.1109/TETC.2022.3186240.
- [58] O. Akbari, M. Kamal, A. Afzali-Kusha and M. Pedram, "Dual-Quality 4:2 Compressors for Utilizing in Dynamic Accuracy Configurable Multipliers," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 25, no. 4, pp. 1352-1361, April 2017, doi: 10.1109/TVLSI.2016.2643003.
- [59] F. Sabetzadeh, M. H. Moaiyeri and M. Ahmadinejad, "A Majority-Based Imprecise Multiplier for Ultra-Efficient Approximate Image Multiplication", *IEEE Transactions on Circuits and Systems-I: Regular Papers*, vol. 66, no. 11, pp. 4200-4208, 2019
- [60] F. Sabetzadeh, M. H. Moaiyeri and M. Ahmadinejad, "An Ultra-Efficient Approximate Multiplier With Error Compensation for Error-Resilient Applications," in *IEEE Transactions on Circuits and Systems*

II: Express Briefs, vol. 70, no. 2, pp. 776-780, Feb. 2023, doi: 10.1109/TCSII.2022.3215065.

- [61] A. G. M. Strollo, E. Napoli, D. De Caro, N. Petra and G. D. Meo, "Comparison and Extension of Approximate 4-2 Compressors for Low-Power Approximate Multipliers," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 67, no. 9, pp. 3021-3034, Sept. 2020, 10.1109/TCSI.2020.2988353.
- [62] E. Zacharelos, C. Scognamiglio, E. Napoli, and D. Gragnaniello https://github.com/dgragnaniello/on-chip_spike_detection.



Ettore Napoli (M'15–SM'15) received the Ph.D. degree in electronic engineering in 1999, the Electronic Engineering degree (Hons.) in 1995, and the Physics degree (Hons.) in 2009.

He was a Research Associate with the Engineering Department, University of Cambridge, U.K., in 2004. Full professor for the University of Napoli Federico II in 2020. Full Professor with the University of Salerno since 2021. He has authored or coauthored more than a hundred articles

published in international journals and conferences. His research interests include VLSI circuit design and modeling and design of power semiconductor devices.



Efstratios Zacharelos (M'24) received the B.Sc. and M.Sc. degrees from the Aristotle University of Thessaloniki, Greece, in physics (2016) and electronic physics (2019), respectively. He received the Ph.D. degree in electronic engineering at Federico II University, Naples, Italy (2023). He is the co-author of nine papers and his main research interests include real-time resampling, approximate computing, and health monitoring devices.



Antonio G. M. Strollo (M'05-SM'06) received the M.S. degree (Hons.) and the PhD degree in Electronic Engineering from the University of Napoli Federico II, Italy. From 2002 he is full professor at the same University, where he has been the head of the Department of Electronic and Telecommunication Engineering from 2005 to 2008. He has published more than 140 papers on international journals and conferences. His current research interest is design and analysis of digital VLSI circuits. Associate Editor for *IEEE Trans. on Circuits and Systems-I* (2009 to 2012); currently he is Associate Editor for *Integration, the VLSI Journal*.



Gennaro Di Meo received the M.S degree (cum laude) in Electronic Engineering and the Ph.D. degree in Information Technology and Electrical Engineering from the University of Naples Federico II, Italy, in 2018 and 2022, respectively. Currently, he is a researcher at the Department of Information Technology and Electrical Engineering at the same university. His research interests include design of digital VLSI circuits for telecommunications, LMS filters and approximate computing.